Course Notes

Lecture 04 — Bluetooth and Bluetooth Low Energy

Communication Technologies for the Internet of Things

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Text to support independent study

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How to Use These Notes

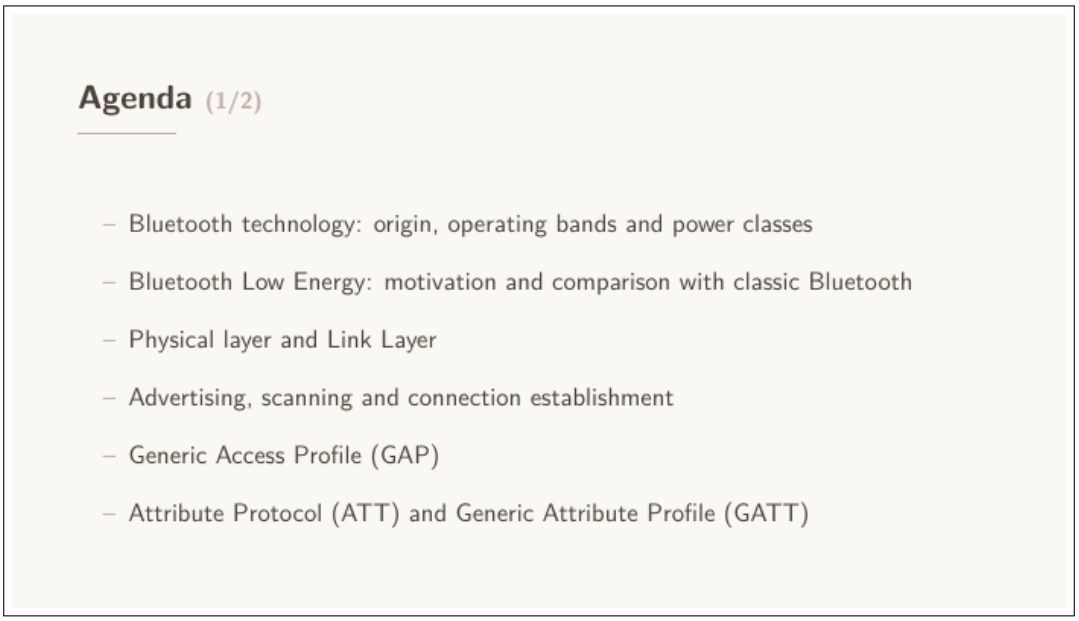
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

1. Commentary on the Slides

Slide 2. Agenda (1/2)



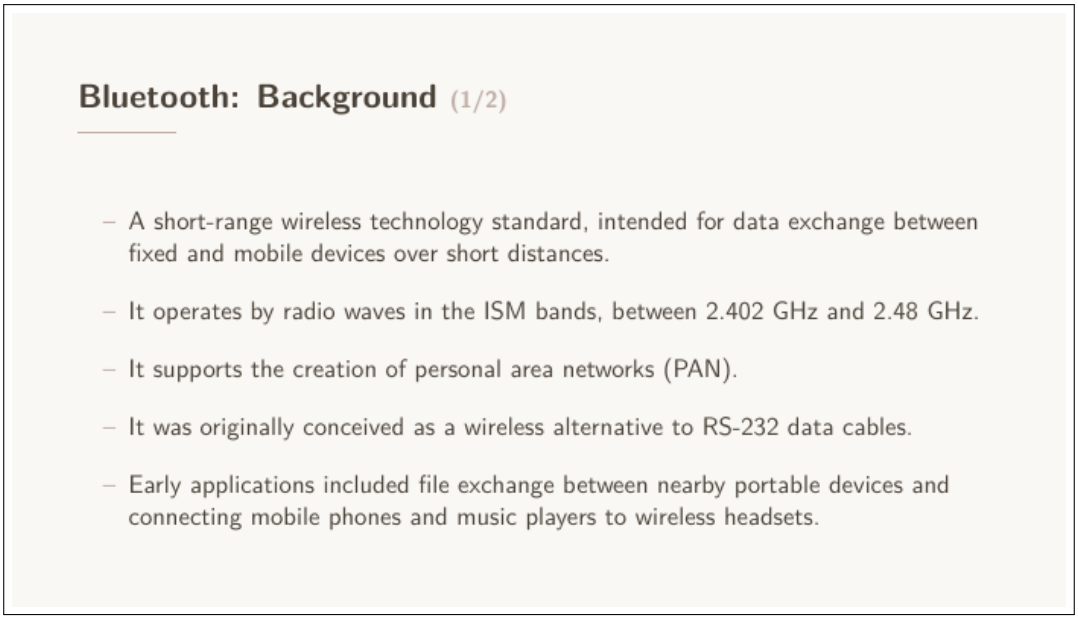
The path through the lecture. The lecture is split into two unequal halves. The first, short, covers classic Bluetooth and justifies the appearance of BLE. The second, long, walks the BLE stack from the bottom up, from the physical layer to the application profiles. This reading order is worth retaining, because each layer only makes sense once the previous one is understood. Whoever tries to study GATT without mastering the link layer will find it difficult.

Slide 3. Agenda (2/2)



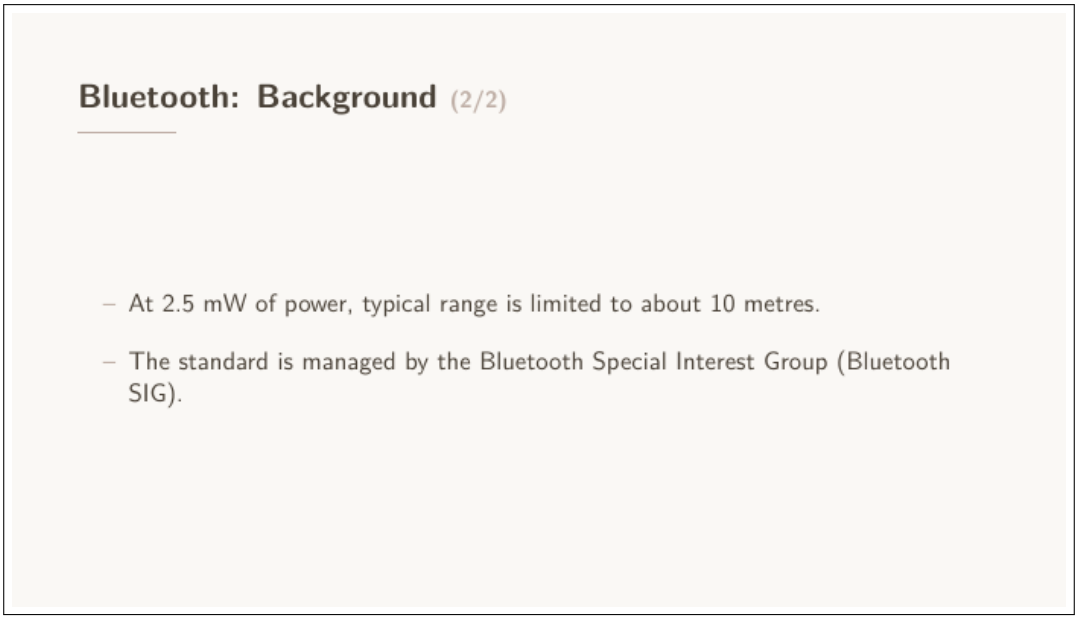
Security and tools. The last two topics close the lecture on the practical side. Security is not an appendix: the interface limitations of sensors condition the pairing method available, and therefore the level of protection achievable. The management tools directly prepare the lab session.

Slide 4. Bluetooth: Background (1/2)



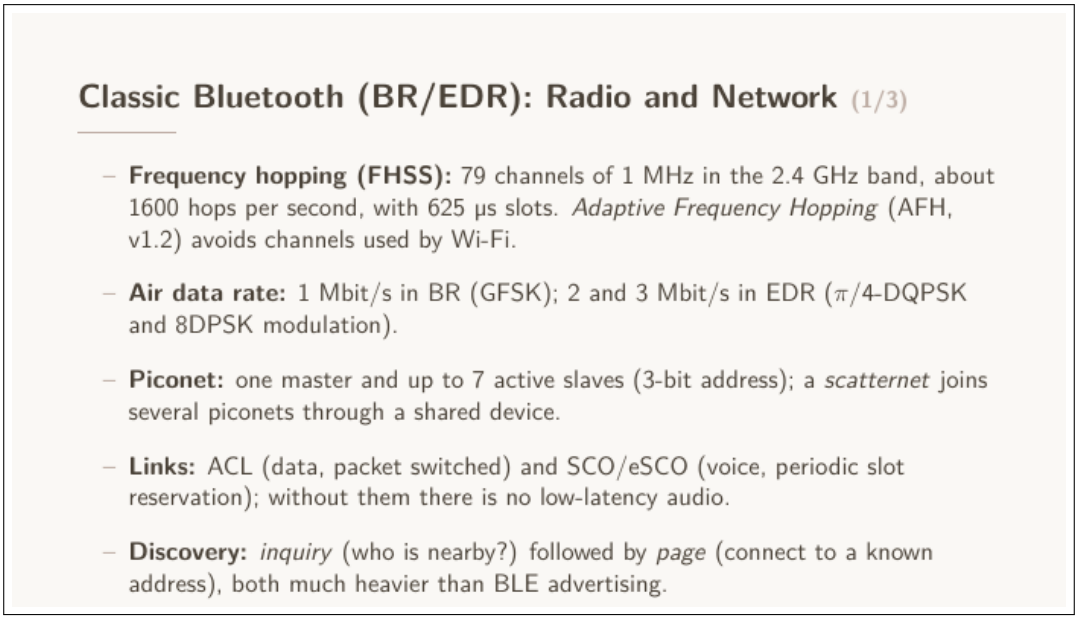
What Bluetooth is and why it was created. Bluetooth was born as a wireless replacement for RS-232 data cables. This origin explains many of its characteristics, among them the short range and the orientation towards connections between two nearby devices. It operates in the ISM bands between 2.402 and 2.48 GHz, an unlicensed band available worldwide, which it shares with Wi-Fi and microwave ovens.

Slide 5. Bluetooth: Background (2/2)



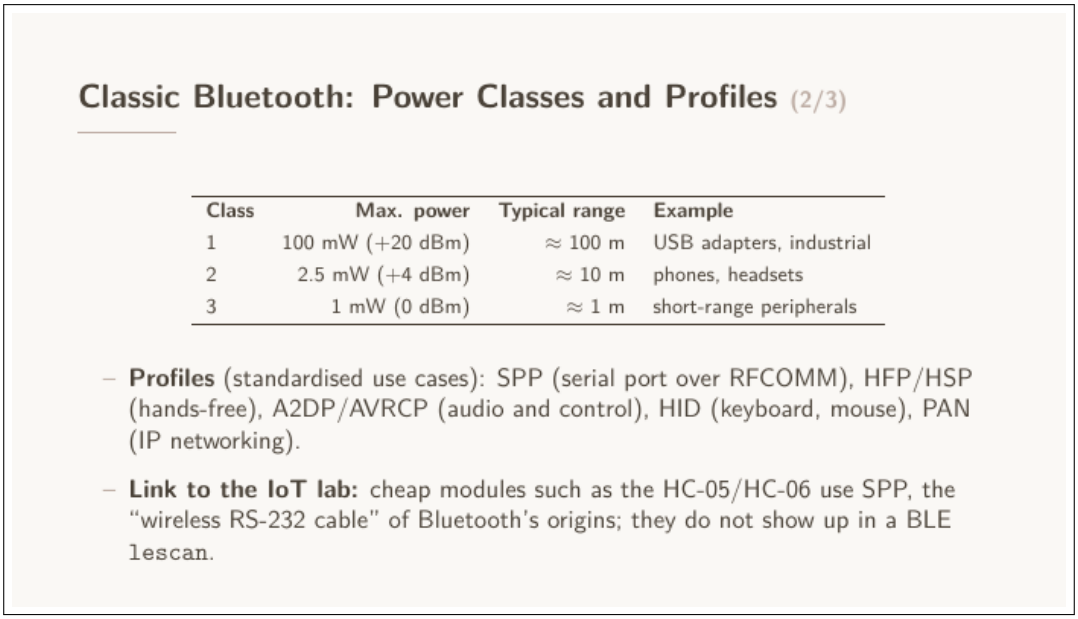
Range and governance of the standard. With a power of 2.5 mW, the typical range is about 10 metres. This value corresponds to the most common power class in consumer equipment. The standard is managed by the Bluetooth Special Interest Group, an industry consortium and not a public standardisation body. This distinction has consequences for access to the specification and for trademark licensing.

Slide 6. Classic Bluetooth (BR/EDR): Radio and Network (1/3)



The classic Bluetooth radio. Classic Bluetooth hops between 79 channels of 1 MHz, about 1600 times a second, and AFH avoids channels occupied by Wi-Fi. A piconet has one master and up to seven active slaves, and ACL and SCO links distinguish data from voice. Discovery by inquiry and page is much heavier than a BLE advertisement, and it is that cost difference that BLE comes to solve.

Slide 7. Classic Bluetooth: Power Classes and Profiles (2/3)



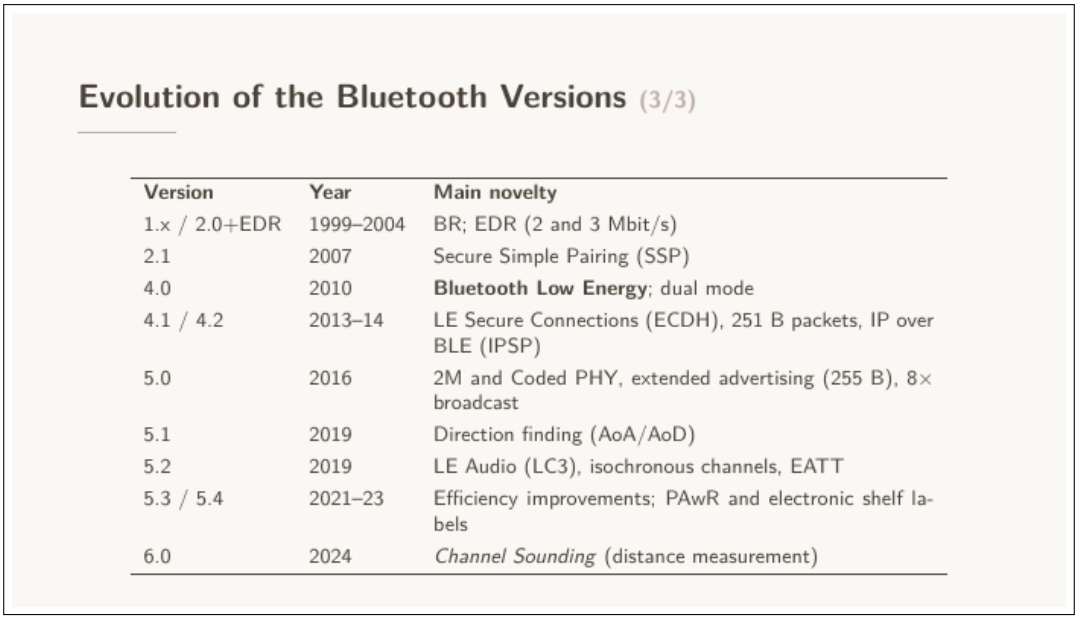
Classic Bluetooth: Power Classes and Profiles (2/3)

Class	Max. power	Typical range	Example
1	100 mW (+20 dBm)	≈ 100 m	USB adapters, industrial
2	2.5 mW (+4 dBm)	≈ 10 m	phones, headsets
3	1 mW (0 dBm)	≈ 1 m	short-range peripherals

- **Profiles** (standardised use cases): SPP (serial port over RFCOMM), HFP/HSP (hands-free), A2DP/AVRCP (audio and control), HID (keyboard, mouse), PAN (IP networking).
- **Link to the IoT lab:** cheap modules such as the HC-05/HC-06 use SPP, the "wireless RS-232 cable" of Bluetooth's origins; they do not show up in a BLE `lescan`.

Power classes and profiles. Profiles standardise use cases, such as the SPP serial port or A2DP audio. In the lab, cheap modules such as the HC-05 use SPP, the wireless RS-232 cable of Bluetooth's origin, and so they do not appear in a BLE scan.

Slide 8. Evolution of the Bluetooth Versions (3/3)

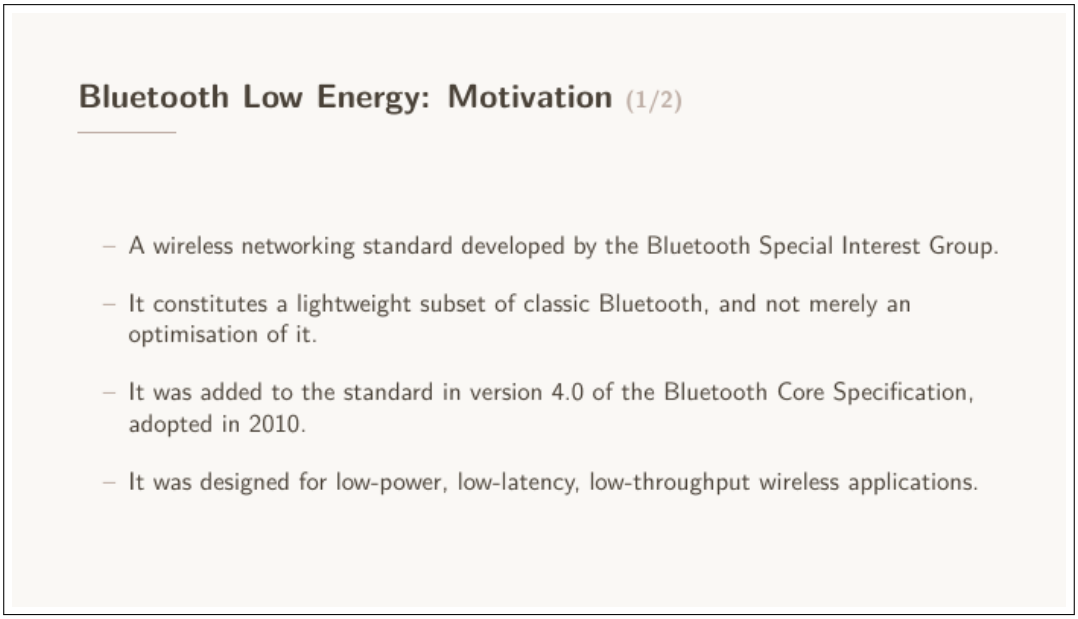


Evolution of the Bluetooth Versions (3/3)

Version	Year	Main novelty
1.x / 2.0+EDR	1999–2004	BR; EDR (2 and 3 Mbit/s)
2.1	2007	Secure Simple Pairing (SSP)
4.0	2010	**Bluetooth Low Energy**; dual mode
4.1 / 4.2	2013–14	LE Secure Connections (ECDH), 251 B packets, IP over BLE (IPSP)
5.0	2016	2M and Coded PHY, extended advertising (255 B), 8× broadcast
5.1	2019	Direction finding (AoA/AoD)
5.2	2019	LE Audio (LC3), isochronous channels, EATT
5.3 / 5.4	2021–23	Efficiency improvements; PAwR and electronic shelf labels
6.0	2024	*Channel Sounding* (distance measurement)

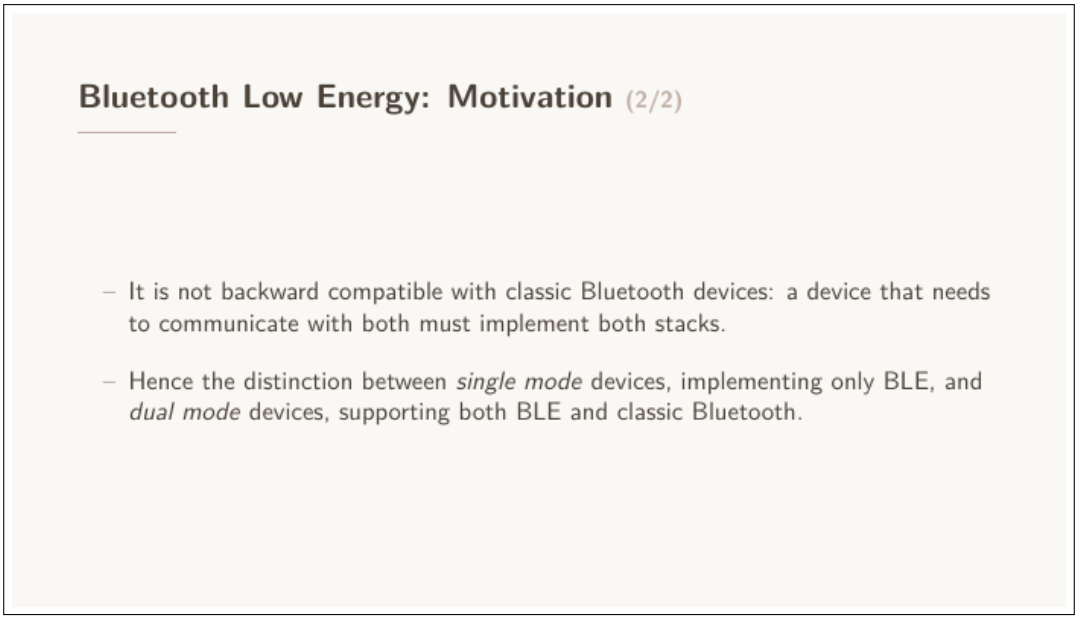
Version evolution. The table shows the path from 1999 to Bluetooth 6.0. The milestones to retain are 4.0 (BLE), 4.2 (Secure Connections and 251-byte packets), 5.0 (long-range PHY and extended advertising) and 6.0 (Channel Sounding). Check: in which version did BLE appear?

Slide 9. Bluetooth Low Energy: Motivation (1/2)



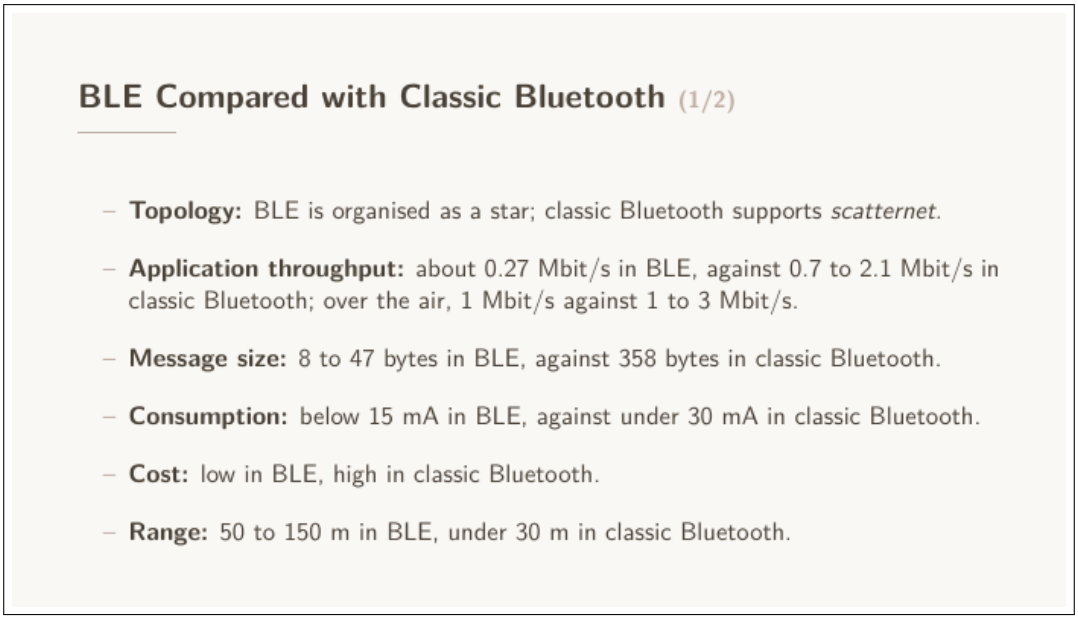
Why BLE is not an optimisation of Bluetooth. This is the most important point of the first half of the lecture, and the one most often misunderstood. BLE is a lightweight subset designed from scratch, and not an improved version of classic Bluetooth. It was added to the standard in version 4.0, adopted in 2010, with a distinct goal: low consumption, low latency and low data rate.

Slide 10. Bluetooth Low Energy: Motivation (2/2)



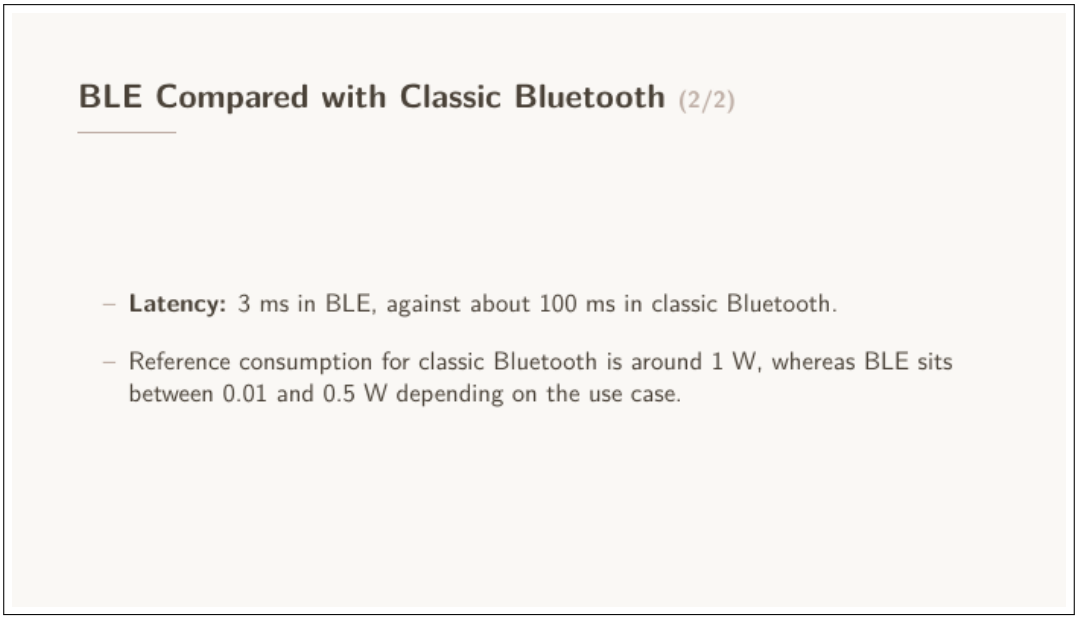
The practical consequence of incompatibility. With no backward compatibility, a device that needs to talk to both has to implement two complete stacks. Hence the commercial distinction between single-mode devices, which only implement BLE, and dual-mode ones, which support both. When choosing a module for a project, this is the first decision to make, and it has a direct impact on cost.

Slide 11. BLE Compared with Classic Bluetooth (1/2)



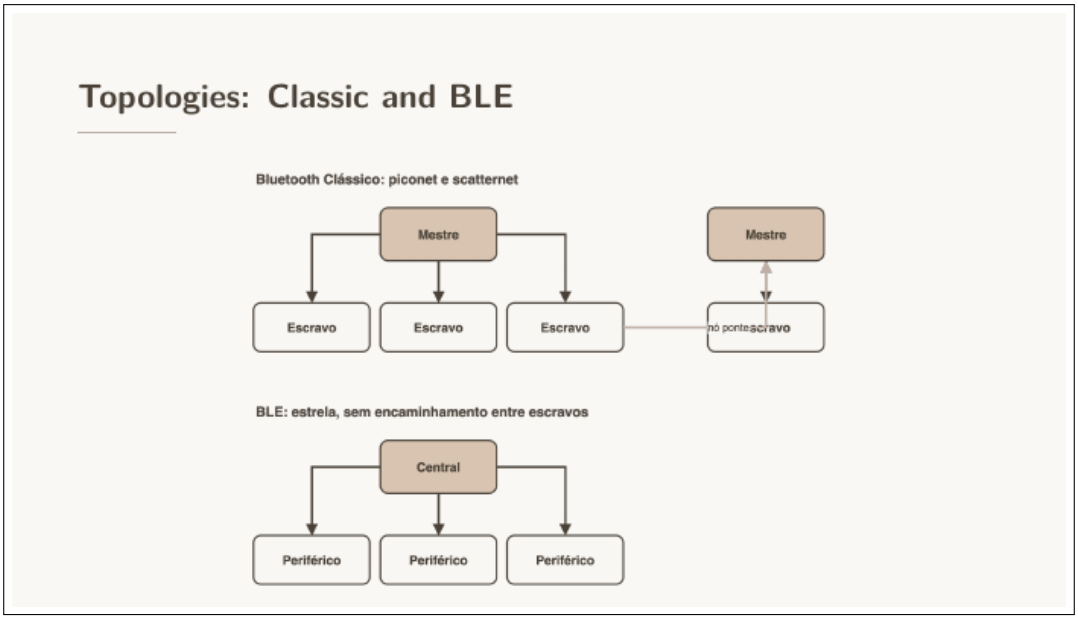
The comparison that justifies the existence of BLE. The numbers on this slide should be read together and not in isolation. BLE loses in application data rate, about 0.27 Mbit/s against 0.7 to 2.1 Mbit/s, and loses a lot in message size, 8 to 47 bytes against 358 bytes. In return it wins in consumption, in cost and, surprisingly, in range: 50 to 150 metres against less than 30 metres. The range gain comes from the higher modulation index, covered later.

Slide 12. BLE Compared with Classic Bluetooth (2/2)



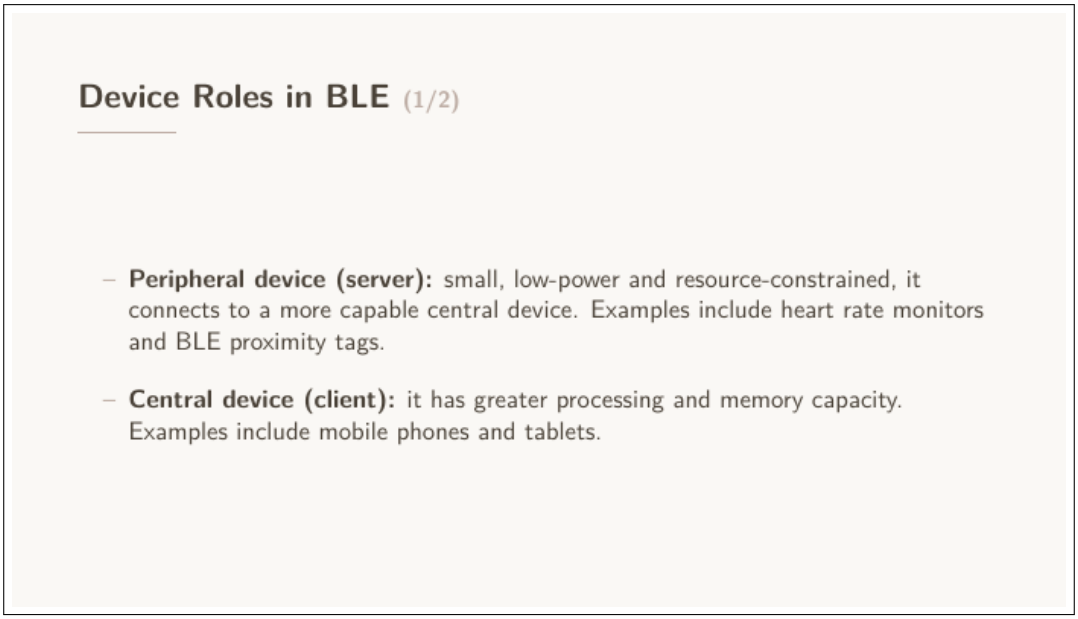
Latency and the order of magnitude of consumption. The latency of 3 ms against about 100 ms is decisive in control applications. The reference consumption, about 1 W for classic against 0.01 to 0.5 W for BLE, is a difference of one to two orders of magnitude. It is this difference that allows a sensor to be powered for years by a coin cell.

Slide 13. Topologies: Classic and BLE



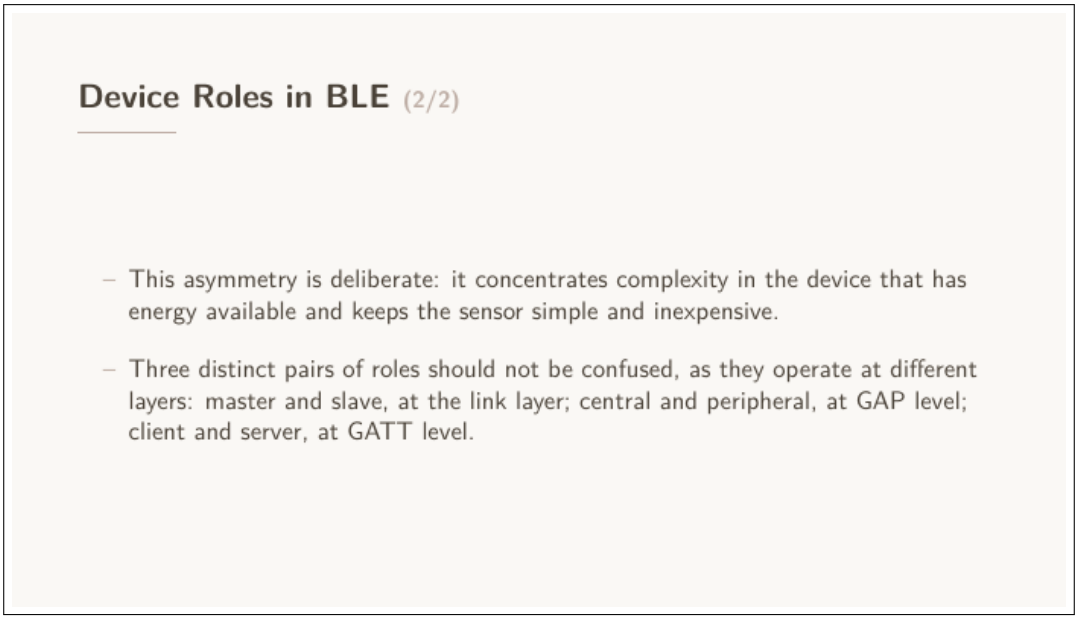
Two families, two topologies. The diagram contrasts the piconet and scatternet of classic Bluetooth with the BLE star. The observation to retain is in the lower half: in BLE there is no routing between slaves. Two peripherals never communicate directly. All messages go through the central device. Whoever compares this lecture with the Zigbee one will find here the structural difference between the two technologies.

Slide 14. Device Roles in BLE (1/2)



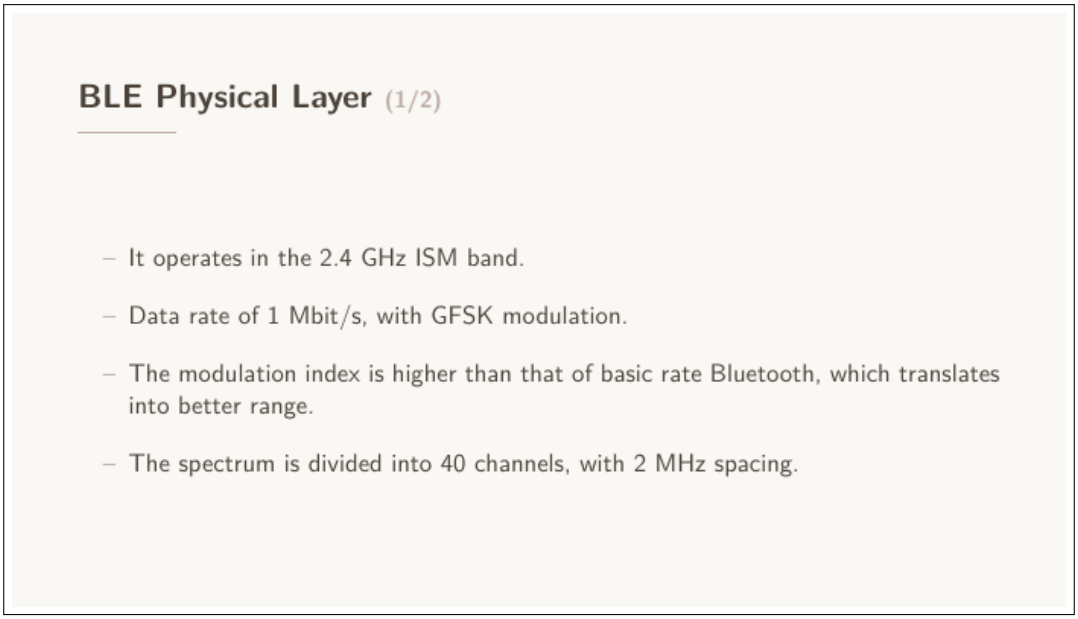
The asymmetry between peripheral and central. The peripheral is small, low-power and resource-limited. The central has more processing and memory. Examples are, respectively, a heart rate monitor and a mobile phone.

Slide 15. Device Roles in BLE (2/2)



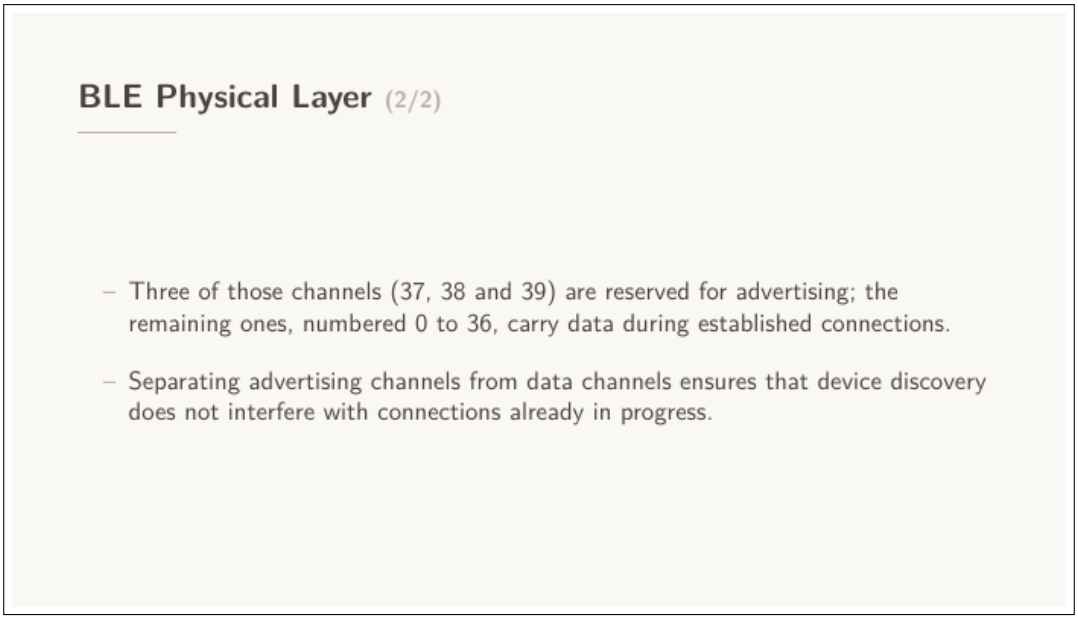
Three pairs of roles that should not be confused. This slide holds exam material. There are three pairs of designations, and they operate at different layers: master and slave at the link layer, central and peripheral at the GAP level, client and server at the GATT level. A device can be both peripheral and GATT server, but the two designations are not synonyms. The asymmetry is deliberate: it concentrates complexity where energy is available.

Slide 16. BLE Physical Layer (1/2)



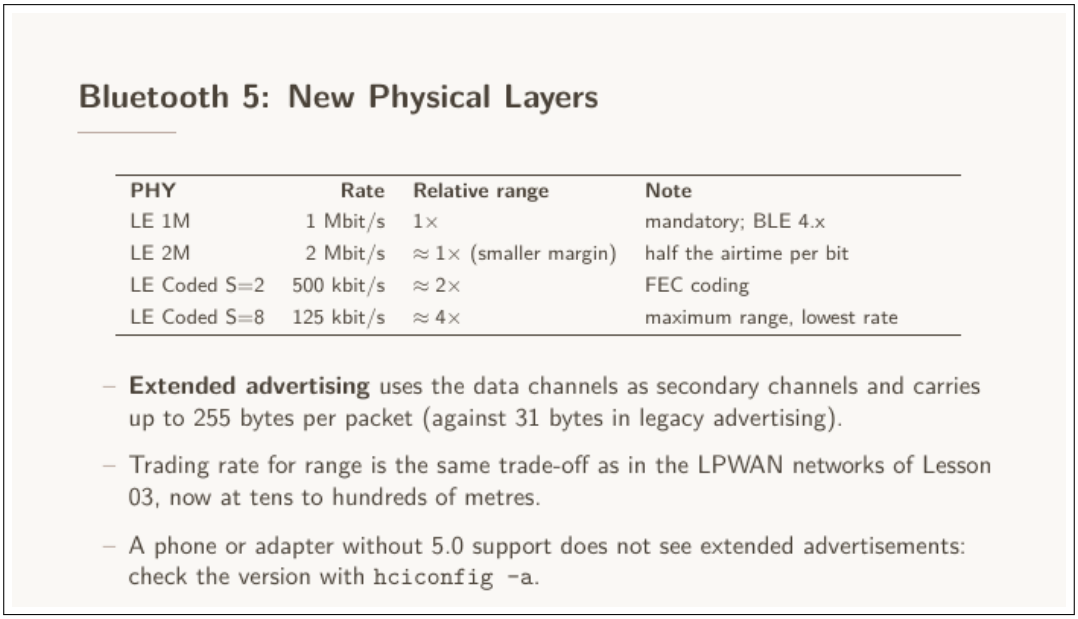
The physical layer and the division of the spectrum. BLE operates in the 2.4 GHz ISM band, at 1 Mbit/s and with GFSK modulation. A modulation index higher than that of basic-rate Bluetooth is the technical reason for the better range noted on slide 8. The spectrum is divided into 40 channels spaced 2 MHz apart.

Slide 17. BLE Physical Layer (2/2)



Why there are dedicated advertising channels. Three channels, 37, 38 and 39, serve exclusively for advertising. The remaining ones, 0 to 36, carry data during connections. The separation is not arbitrary: it guarantees that device discovery, which is permanent and uncoordinated traffic, does not disturb connections already established.

Slide 18. Bluetooth 5: New Physical Layers



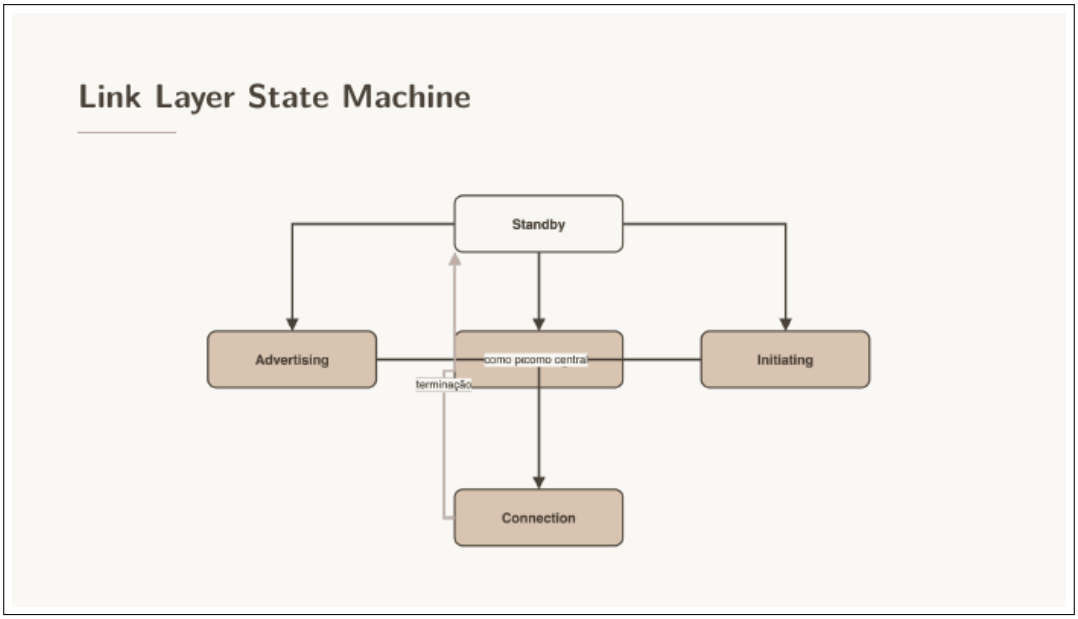
Bluetooth 5: New Physical Layers

PHY	Rate	Relative range	Note
LE 1M	1 Mbit/s	1×	mandatory; BLE 4.x
LE 2M	2 Mbit/s	≈ 1× (smaller margin)	half the airtime per bit
LE Coded S=2	500 kbit/s	≈ 2×	FEC coding
LE Coded S=8	125 kbit/s	≈ 4×	maximum range, lowest rate

- **Extended advertising** uses the data channels as secondary channels and carries up to 255 bytes per packet (against 31 bytes in legacy advertising).
- Trading rate for range is the same trade-off as in the LPWAN networks of Lesson 03, now at tens to hundreds of metres.
- A phone or adapter without 5.0 support does not see extended advertisements: check the version with `hciconfig -a`.

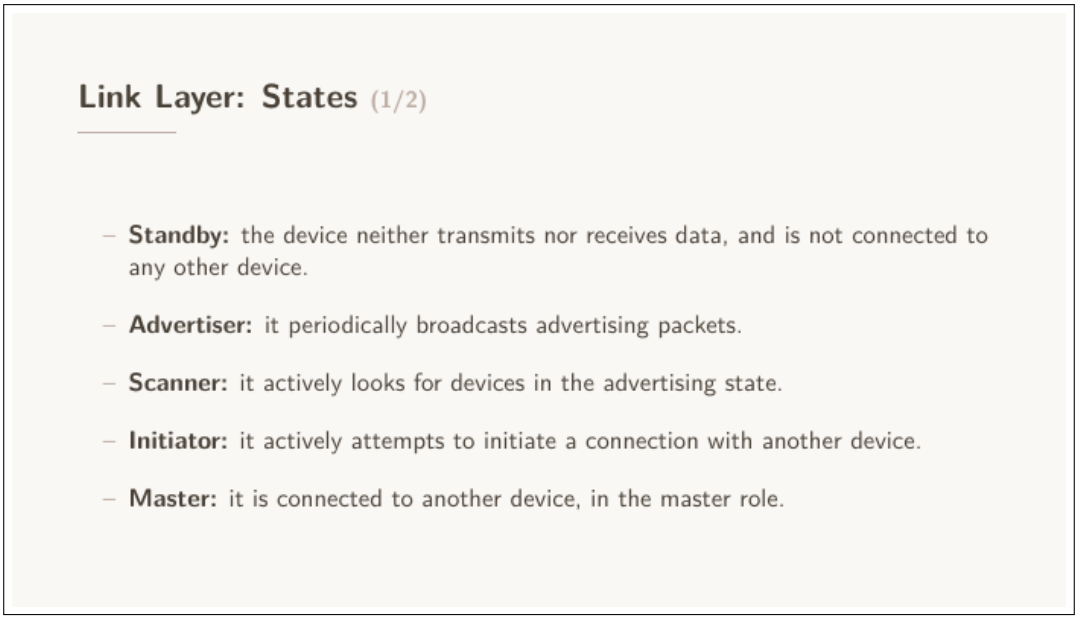
Bluetooth 5: new physical layers. Bluetooth 5 adds a 2 Mbit/s PHY and a coded long-range PHY, trading data rate for range, as in LPWANs. Extended advertisements go from 31 to 255 bytes, but are only seen by adapters with 5.0 support, which is worth checking.

Slide 19. Link Layer State Machine



The state machine in brief. The diagram shows the five main states and the possible transitions. Note that the connected state is reached by two distinct paths: from Advertising, taking the peripheral role, or from Initiating, taking the central one. This diagram is the best revision tool for the whole link layer.

Slide 20. Link Layer: States (1/2)



The states, one by one. Standby corresponds to inactivity. Advertiser broadcasts periodically. Scanner actively looks for those advertising. Initiator tries to establish a connection. Master is the role taken after a connection established from Initiator.

Slide 21. Link Layer: States (2/2)



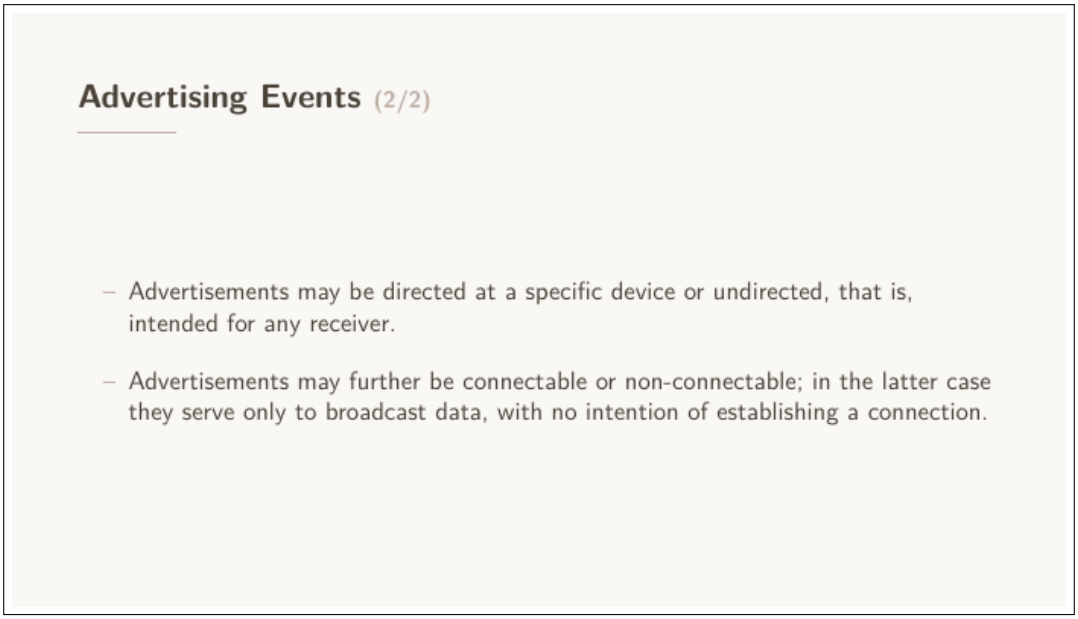
The sixth state and the complete cycle. Slave is the role taken after a connection established from Advertiser. The transitions between the six states describe the whole life cycle of a BLE communication, from discovery to termination.

Slide 22. Advertising Events (1/2)



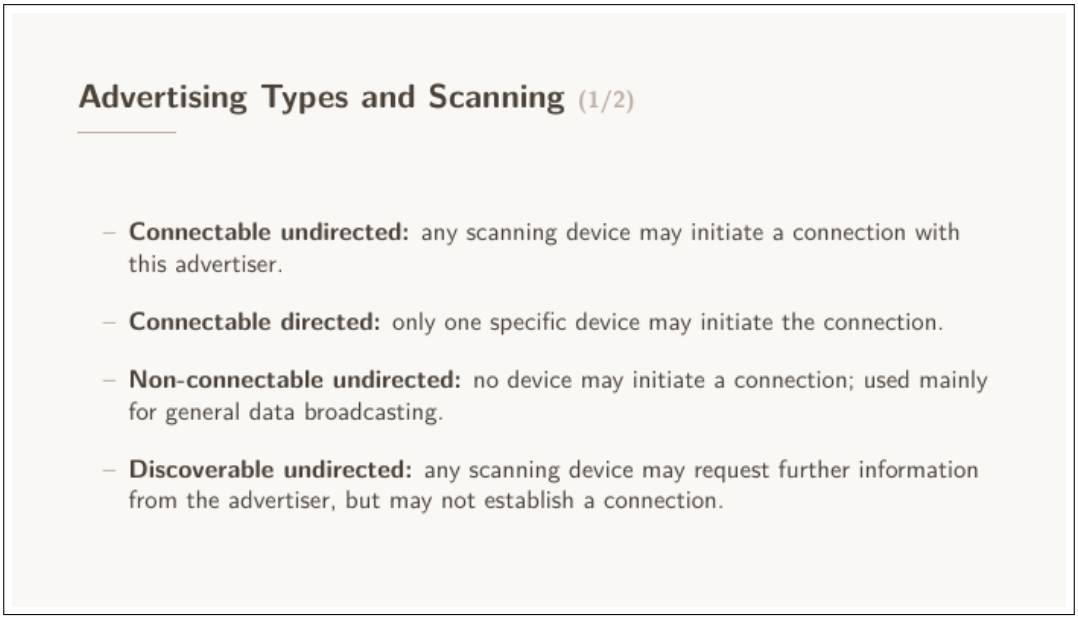
Redundancy in advertising. In an advertising event, one packet is transmitted on each of the three channels. The repetition is not waste: it raises the probability of reception if one of the channels is subject to interference, which in the 2.4 GHz band is frequent.

Slide 23. Advertising Events (2/2)



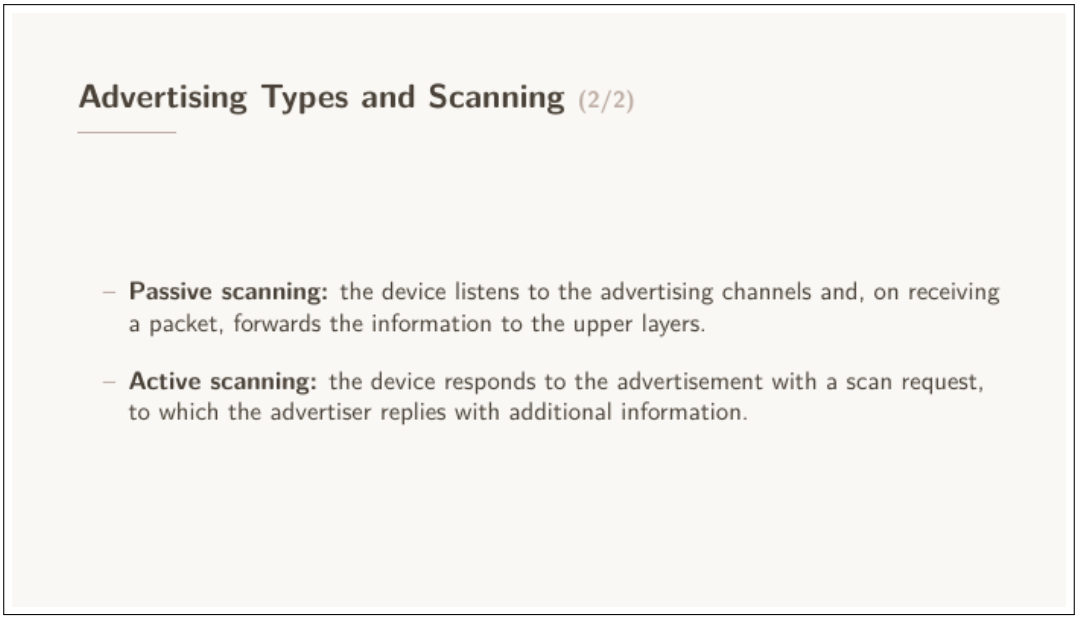
Directed and connectable advertising. There are two independent dimensions of classification. An advertisement can be directed at a specific device or undirected. And it can be connectable or non-connectable. A non-connectable advertisement serves only to broadcast data, with no intention of establishing a connection. It is the mechanism on which beacons are based.

Slide 24. Advertising Types and Scanning (1/2)



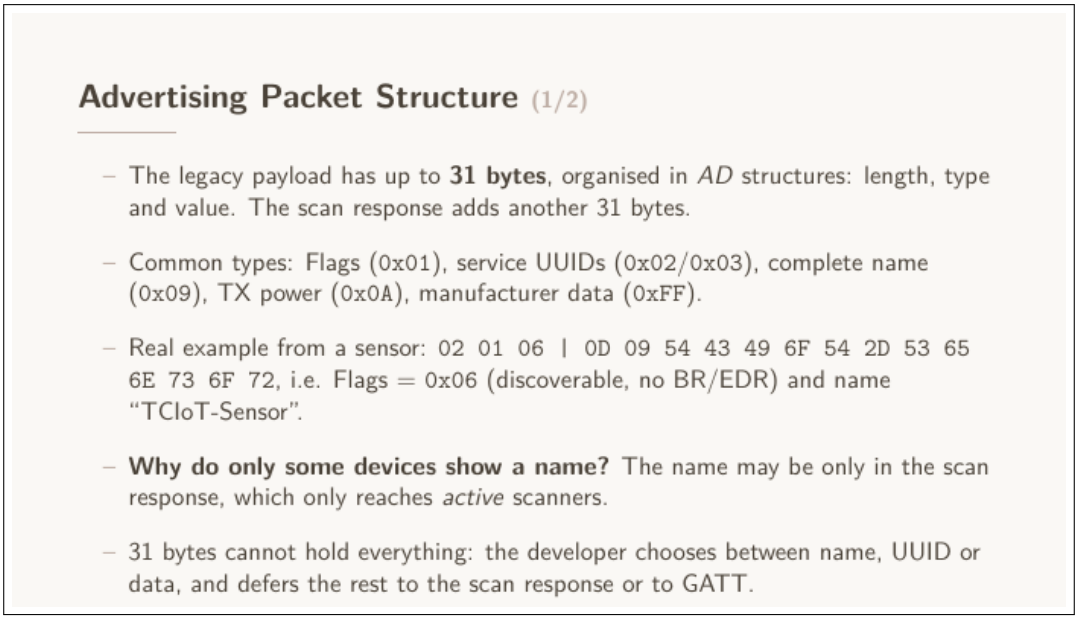
The four combinations. Connectable undirected, connectable directed, non-connectable undirected and discoverable undirected. Knowing how to associate each combination with a use case is more useful than memorising the names.

Slide 25. Advertising Types and Scanning (2/2)



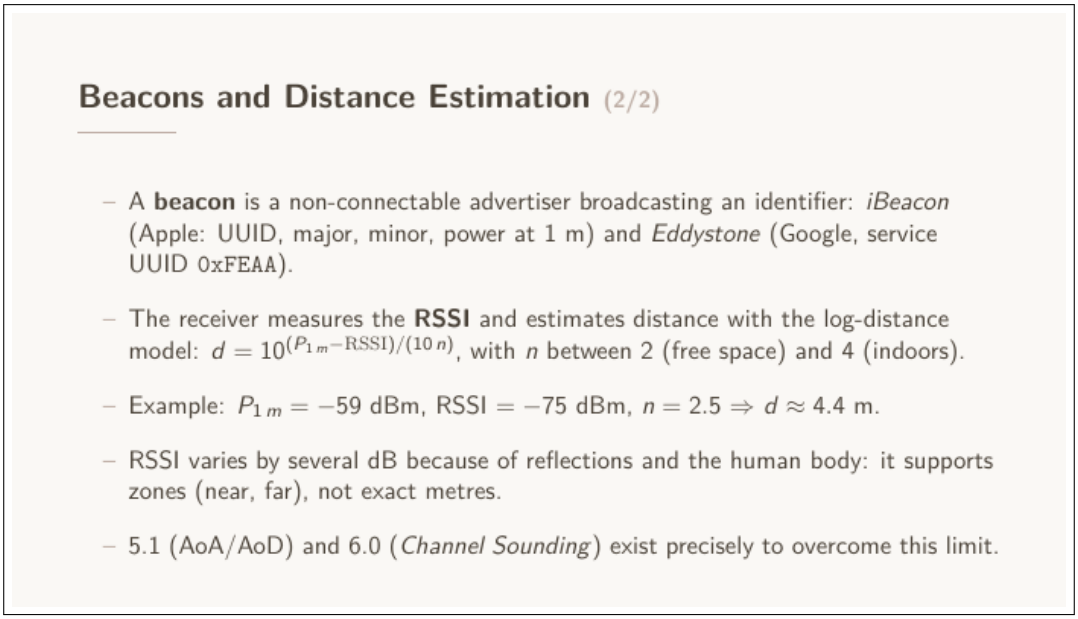
Passive and active scanning. In passive scanning the device just listens. In active scanning it answers the advertisement with a scan request, to which the advertiser answers with additional information. This is why, in the lab session, some devices appear with a name and others only with an address.

Slide 26. Advertising Packet Structure (1/2)



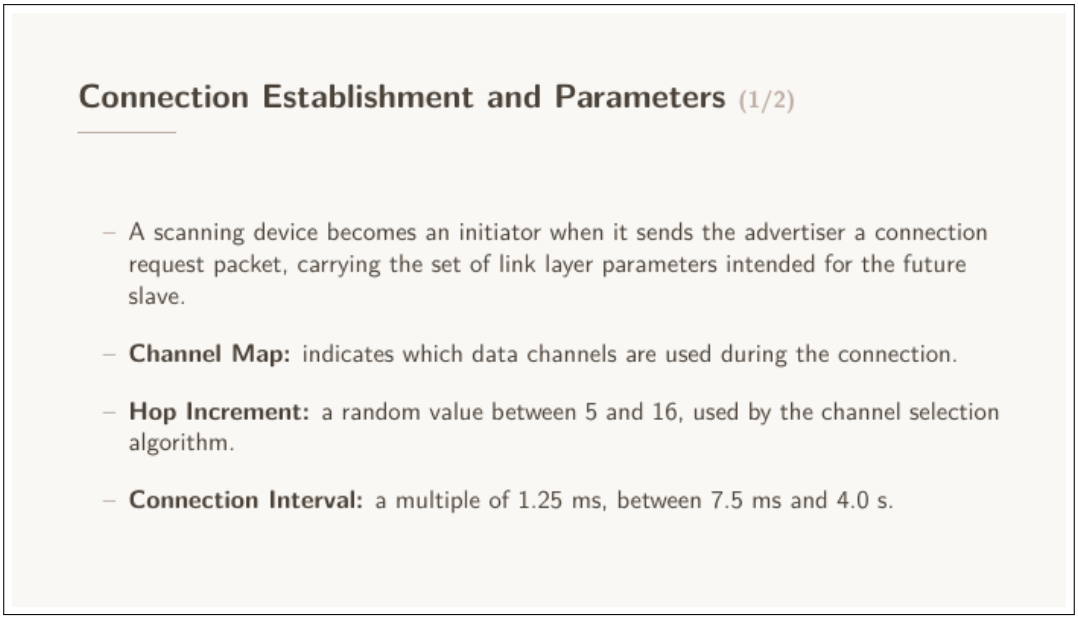
The structure of the advertising packet. The payload of a legacy advertisement has 31 bytes organised in length, type and value structures. The hexadecimal example breaks down into Flags and the complete name. The name may be only in the scan response, which is why only active scanning shows it. With 31 bytes you have to choose what to advertise.

Slide 27. Beacons and Distance Estimation (2/2)



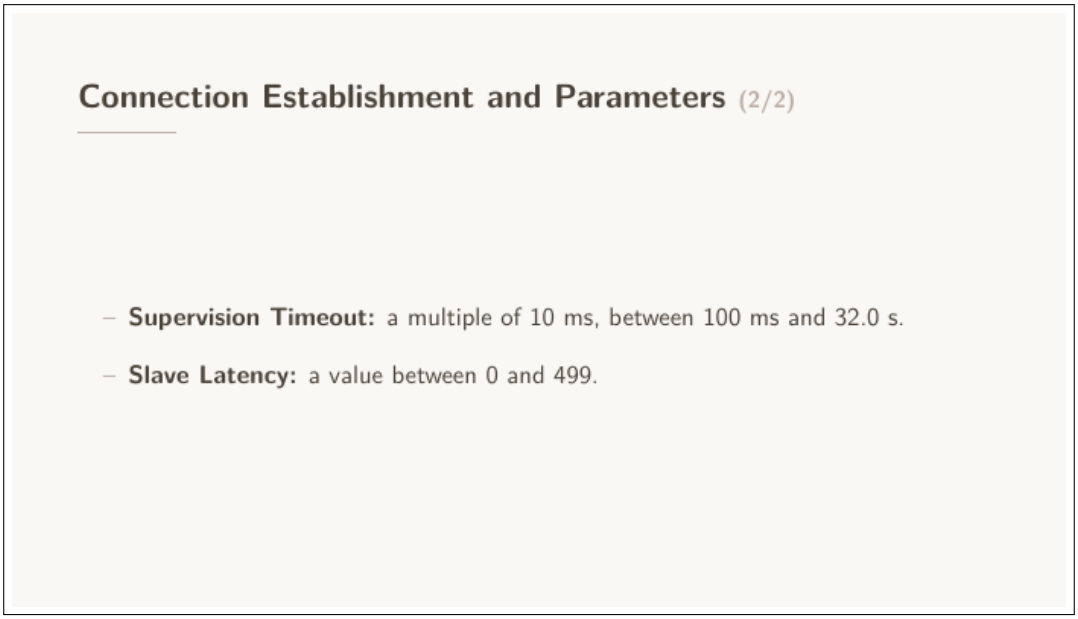
Beacons and distance. A beacon is a non-connectable advertiser. The receiver measures the RSSI and estimates distance with the log-distance model. The example gives about 4.4 metres, but RSSI fluctuates with reflections and the human body, so it serves for zones and not exact metres. Check: redo the calculation with n equal to 2.

Slide 28. Connection Establishment and Parameters (1/2)



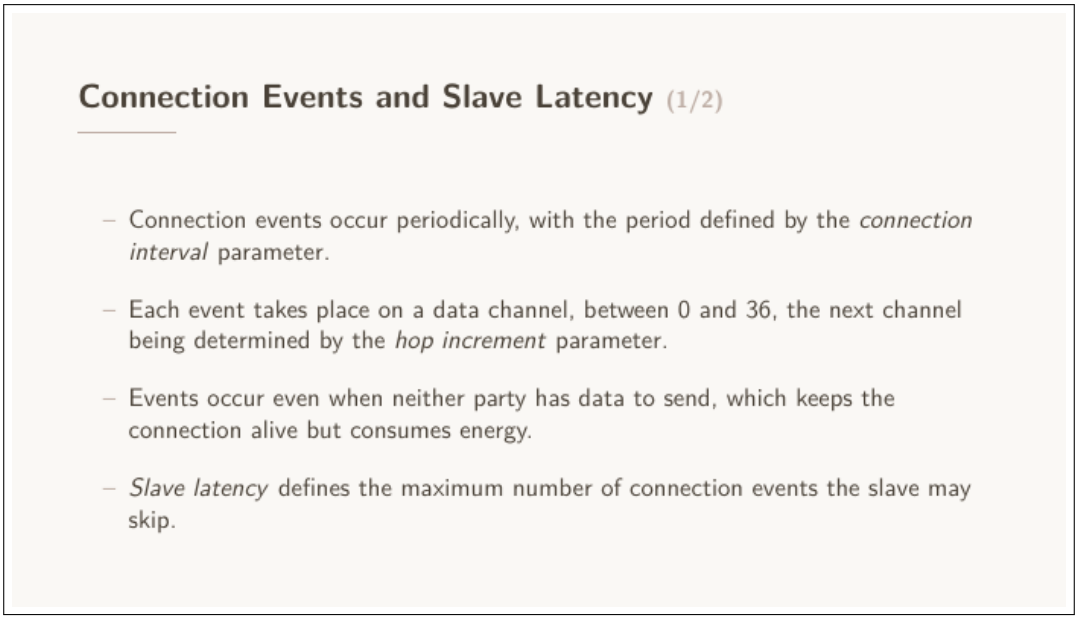
The parameters negotiated in the connection request. On sending the connection request, the initiator transmits the parameters that will govern the connection. The channel map says which data channels to use. The hop increment, a random value between 5 and 16, feeds the channel selection algorithm. The connection interval is a multiple of 1.25 ms, between 7.5 ms and 4 seconds.

Slide 29. Connection Establishment and Parameters (2/2)



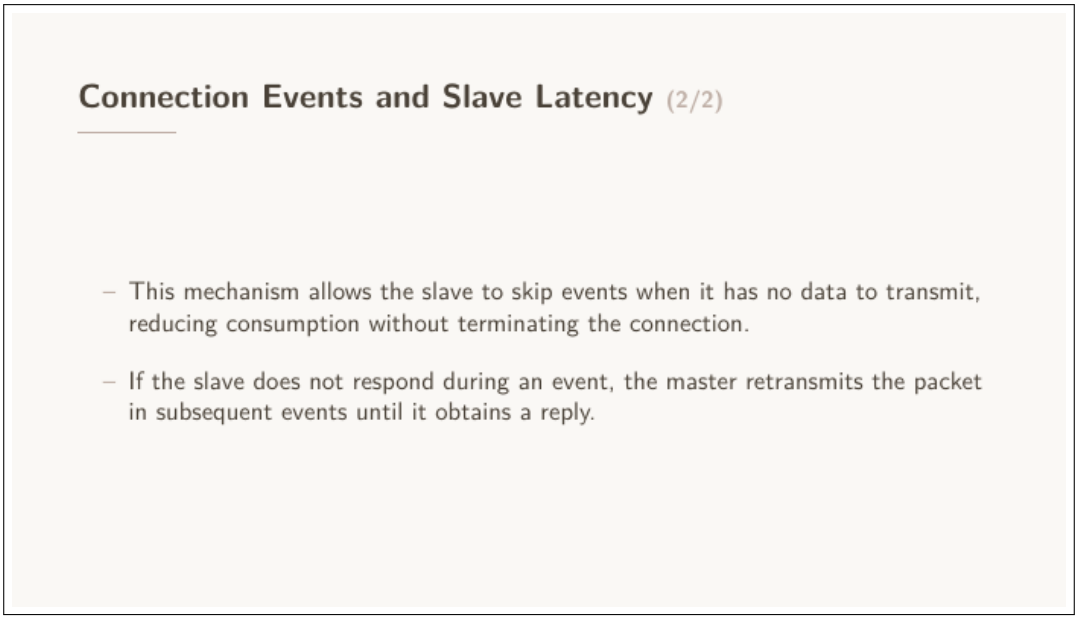
Supervision and slave latency. The supervision timeout is a multiple of 10 ms, between 100 ms and 32 seconds. Slave latency accepts values between 0 and 499. These three parameters, together with the interval, are the subject of the following slides.

Slide 30. Connection Events and Slave Latency (1/2)



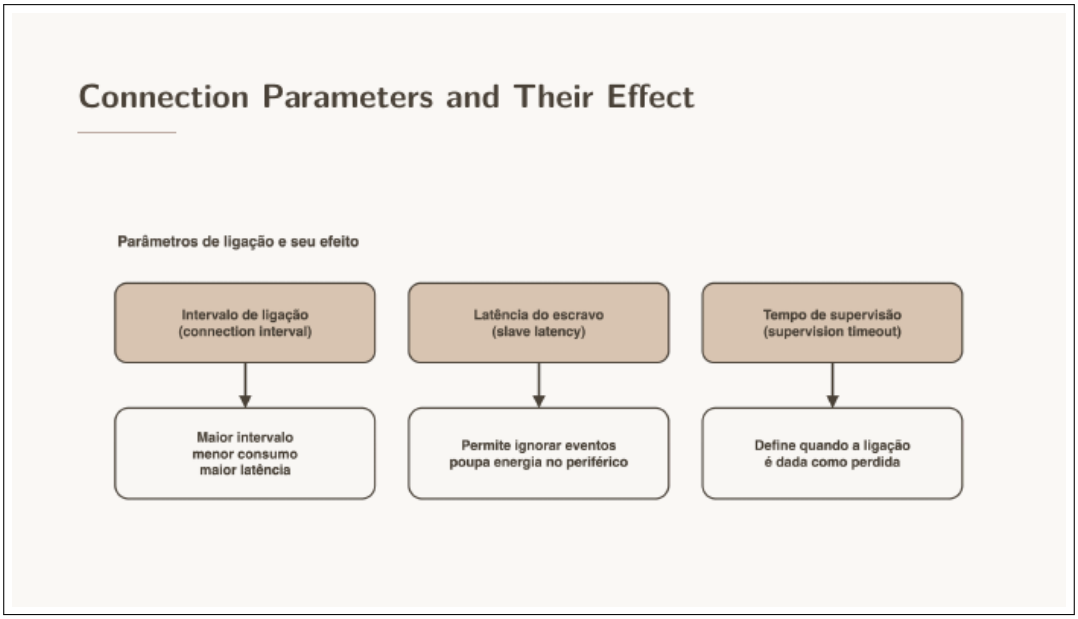
Periodic events and the right to skip. Connection events occur with the period defined by the connection interval, even when neither party has data to send. This upkeep has an energy cost. Slave latency defines how many consecutive events the slave may ignore.

Slide 31. Connection Events and Slave Latency (2/2)



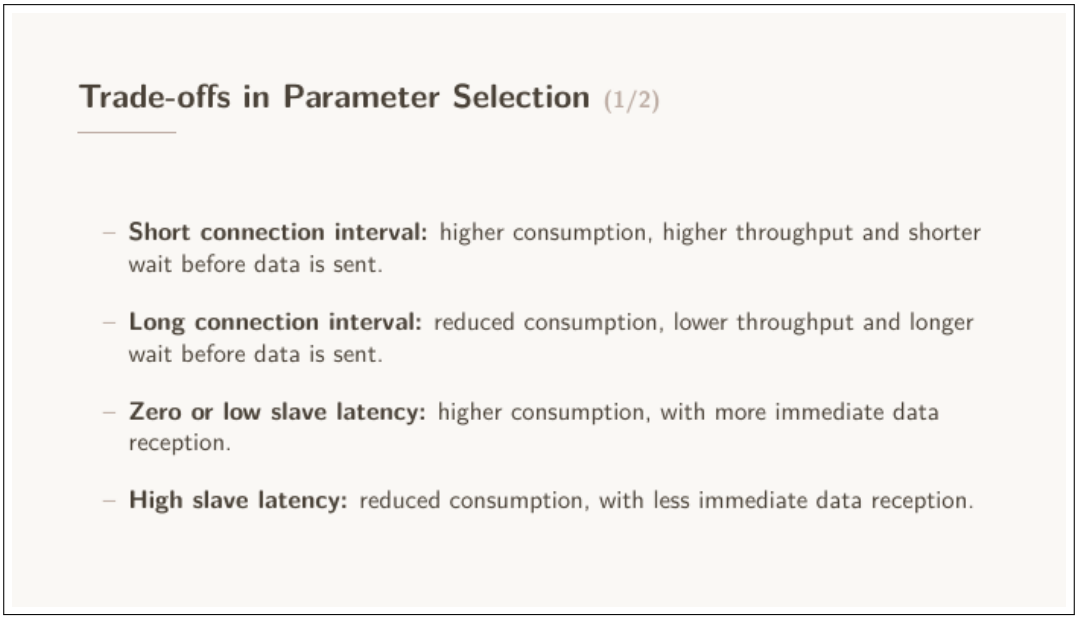
How the mechanism saves energy without breaking the connection. The slave skips events when it has nothing to transmit, reducing consumption without ending the connection. If it does not answer during an event, the master resends in the following ones until it gets a reply.

Slide 32. Connection Parameters and Their Effect



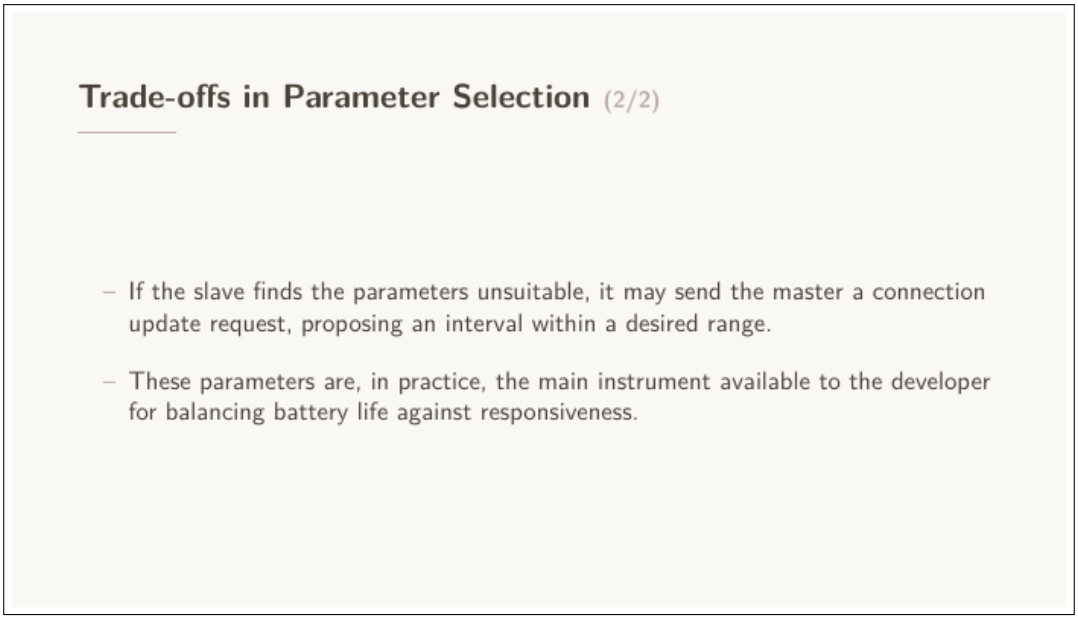
The three parameters and what each controls. The diagram associates each parameter with its effect. A longer interval means lower consumption and higher latency. Slave latency lets the peripheral ignore events and save energy. The supervision timeout defines when the connection is considered lost. This is the information that applies directly when configuring a real device.

Slide 33. Trade-offs in Parameter Selection (1/2)



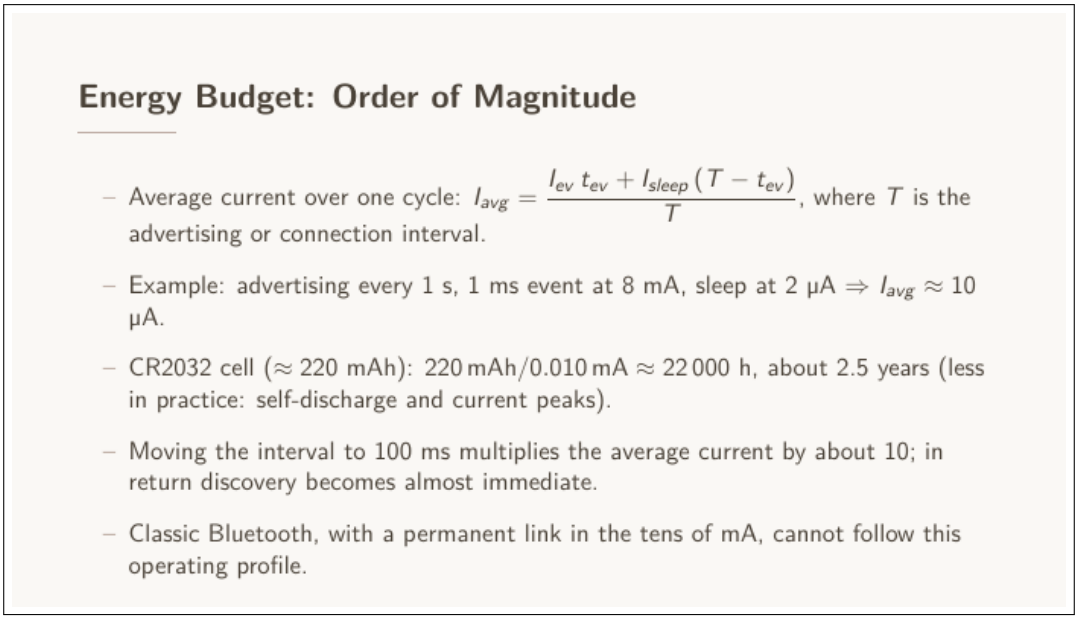
The trade-off, stated in both directions. A short interval means high consumption, higher throughput and less waiting. A long interval means the opposite. Low slave latency means more immediate reception at a higher energy cost. It is worth studying the four lines together, because they describe two variables moving in opposite directions.

Slide 34. Trade-offs in Parameter Selection (2/2)



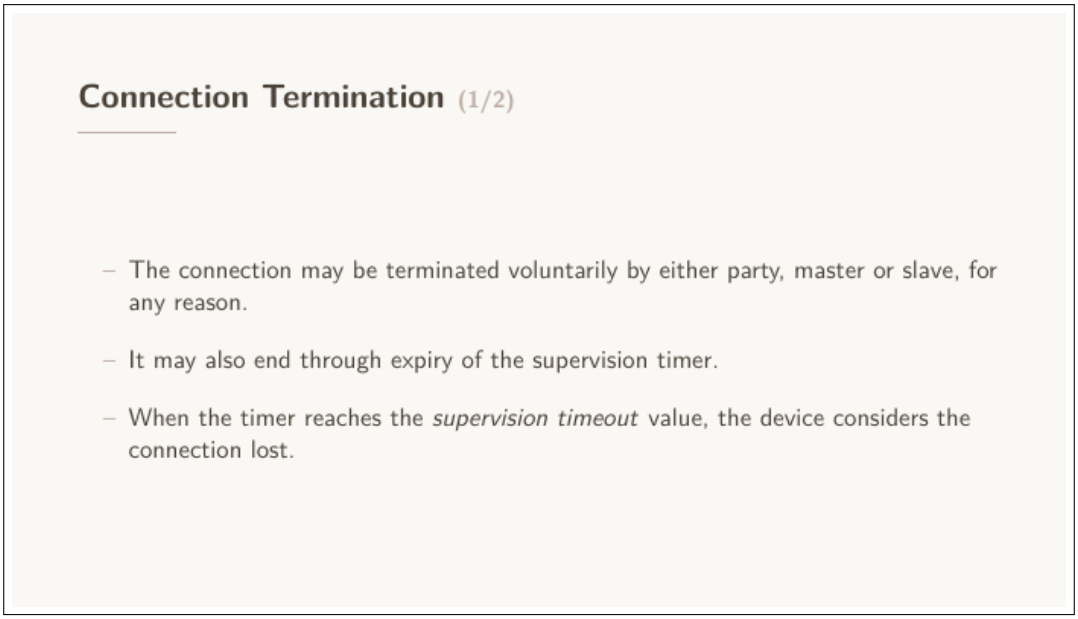
Who decides the parameters. The slave is not obliged to accept what the master proposes. It may send a connection update request, suggesting an interval within a desired range. In practice, these parameters are the programmer's main instrument for balancing autonomy and responsiveness.

Slide 35. Energy Budget: Order of Magnitude



Energy budget. The average current is the weighted average between the event and rest. With an advertisement every second, a 1 ms event at 8 mA and rest at 2 microamps, the average is about 10 microamps, which gives about 2.5 years on a CR2032 cell. Reducing the interval to 100 ms multiplies the current by ten.

Slide 36. Connection Termination (1/2)



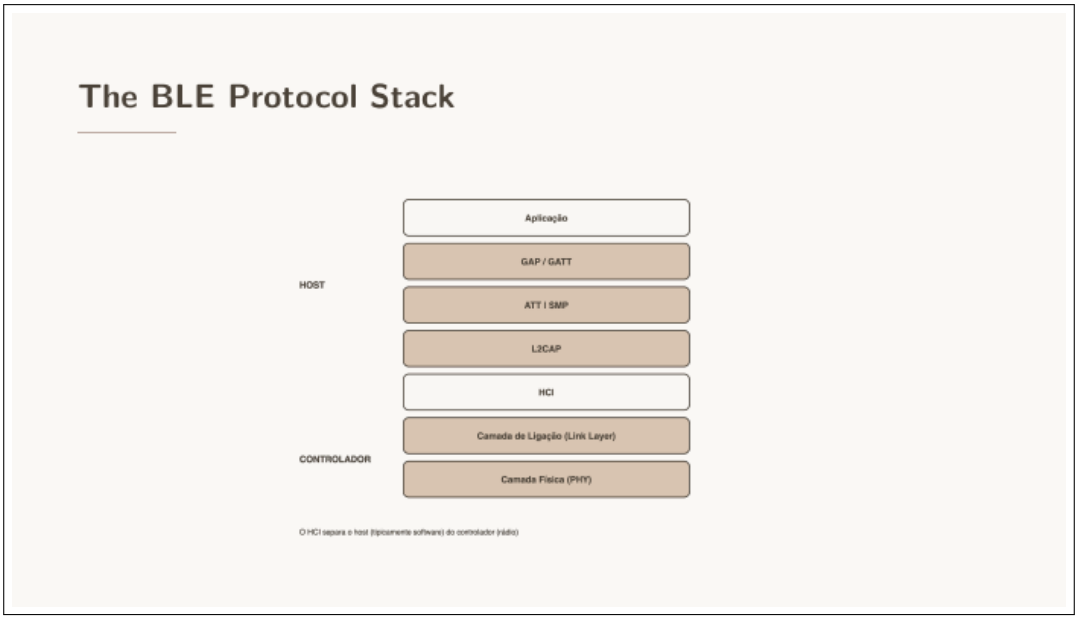
Two ways to end. Termination may be voluntary, at the initiative of either party, or involuntary, by expiry of the supervision timer. In the second case, the device considers the connection lost.

Slide 37. Connection Termination (2/2)



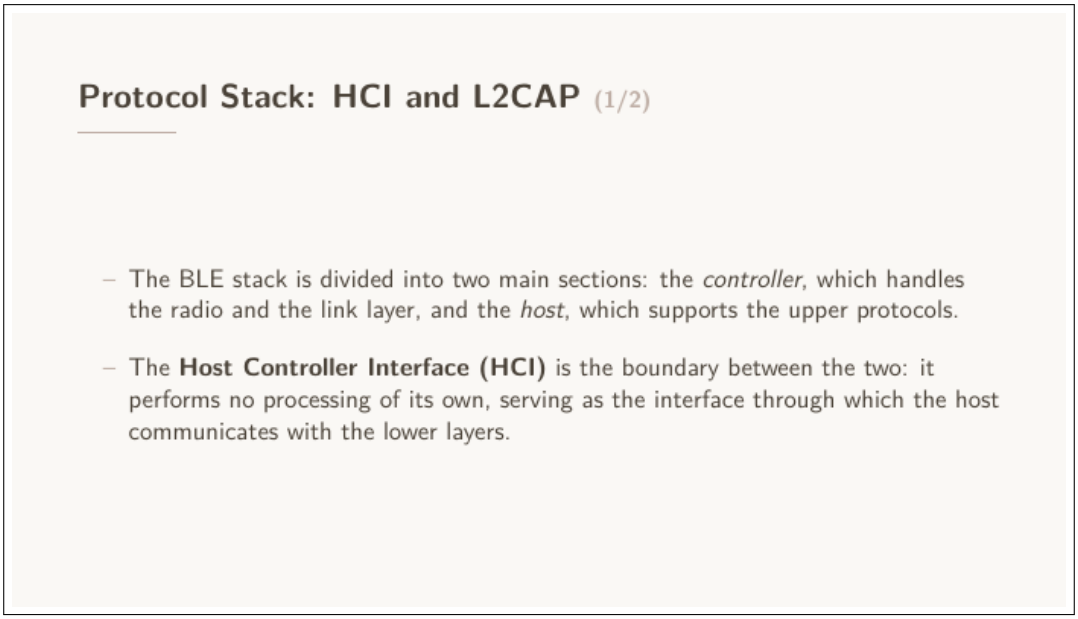
What happens next. Once the connection is lost, the device returns to the advertising, scanning or standby state, depending on its role. The mechanism guarantees that no device is left indefinitely waiting for a peer that has stopped responding. In the lab session this time is measured experimentally.

Slide 38. The BLE Protocol Stack



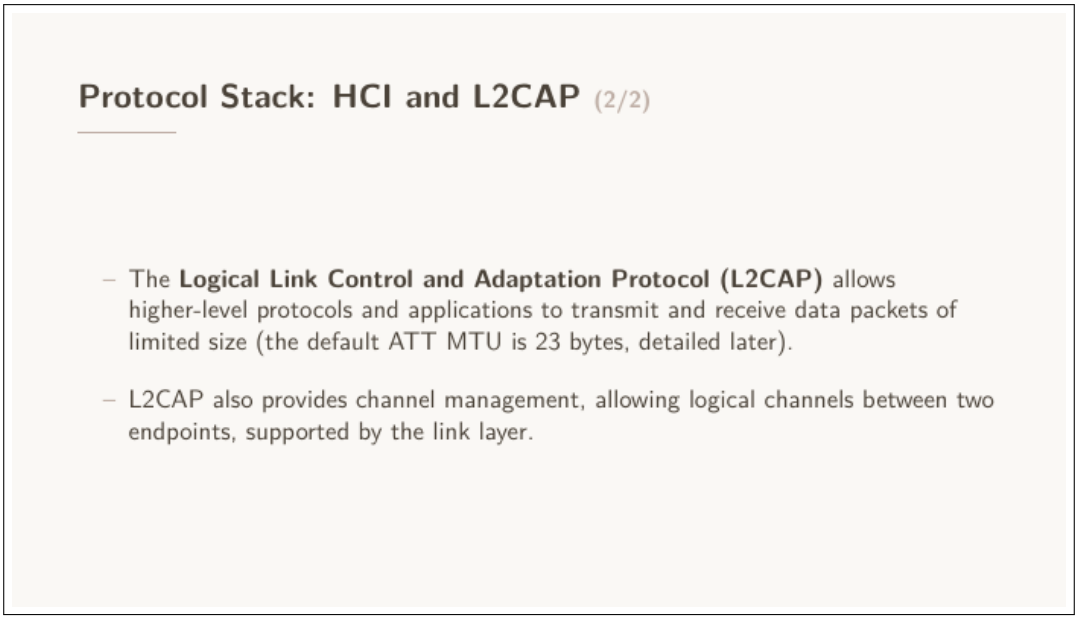
The stack, seen at once. The diagram separates the controller, which handles the radio and the link layer, from the host, which supports the upper protocols. HCI is the boundary between the two. It is worth retaining the relative position of each block, because the following slides cover each in isolation and it is easy to lose sight of the whole.

Slide 39. Protocol Stack: HCI and L2CAP (1/2)



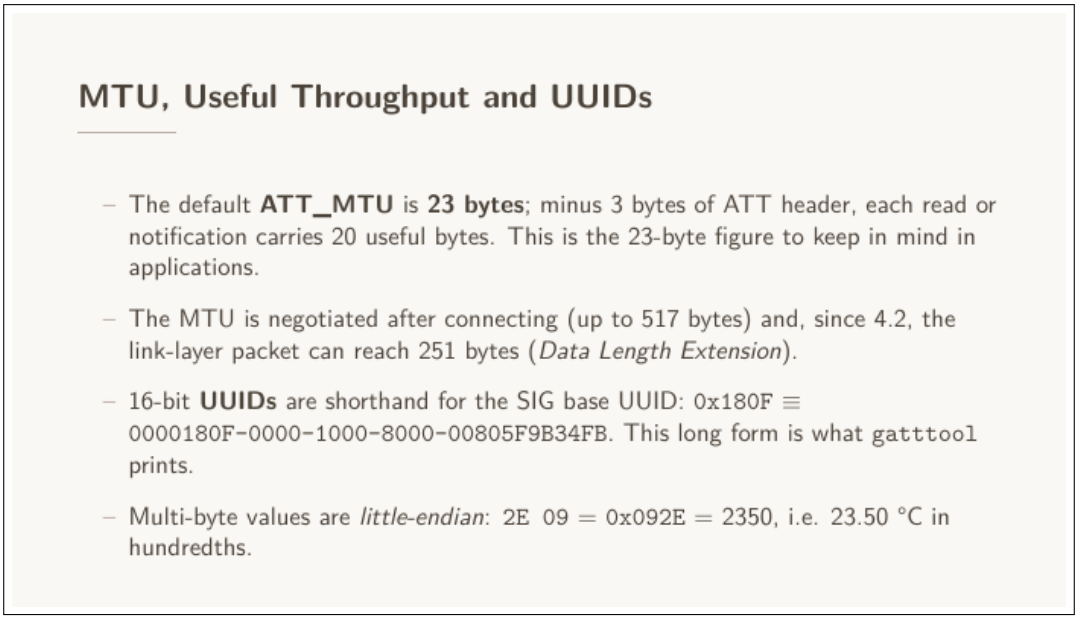
The role of HCI. HCI performs no processing of its own. It is just the interface through which the host communicates with the lower layers. Its existence allows host and controller to sit on different chips, connected by serial port or USB, which is the usual configuration in a computer.

Slide 40. Protocol Stack: HCI and L2CAP (2/2)



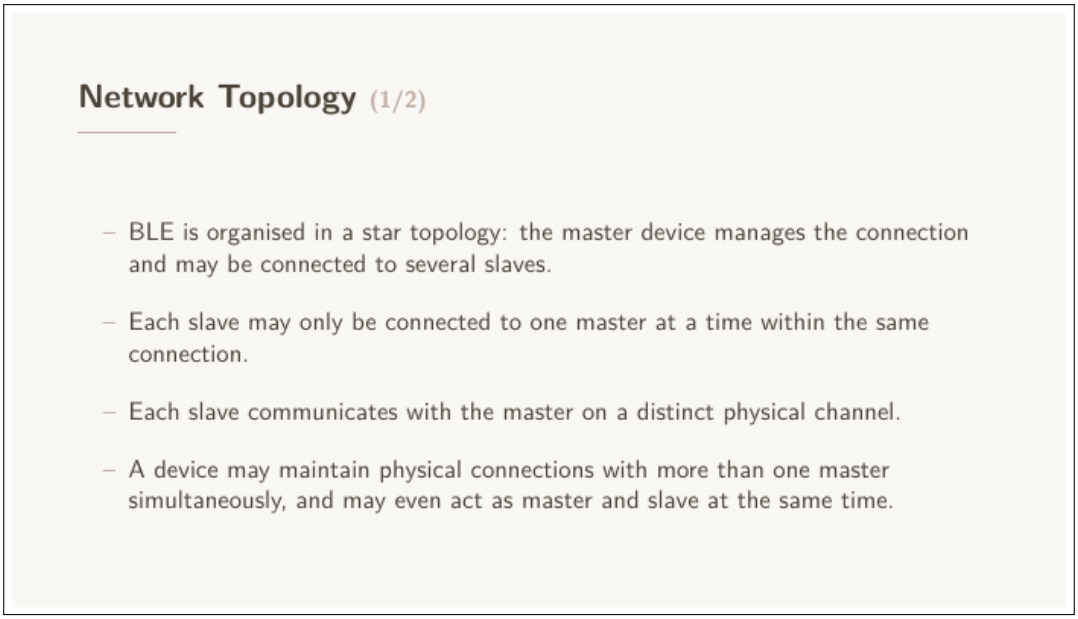
L2CAP and the 23-byte limit. L2CAP lets upper layers transmit packets of up to 23 bytes, and manages logical channels. The 23-byte value explains why transferring any appreciable volume of data over BLE is slow: the information has to be fragmented into many small messages.

Slide 41. MTU, Useful Throughput and UUIDs



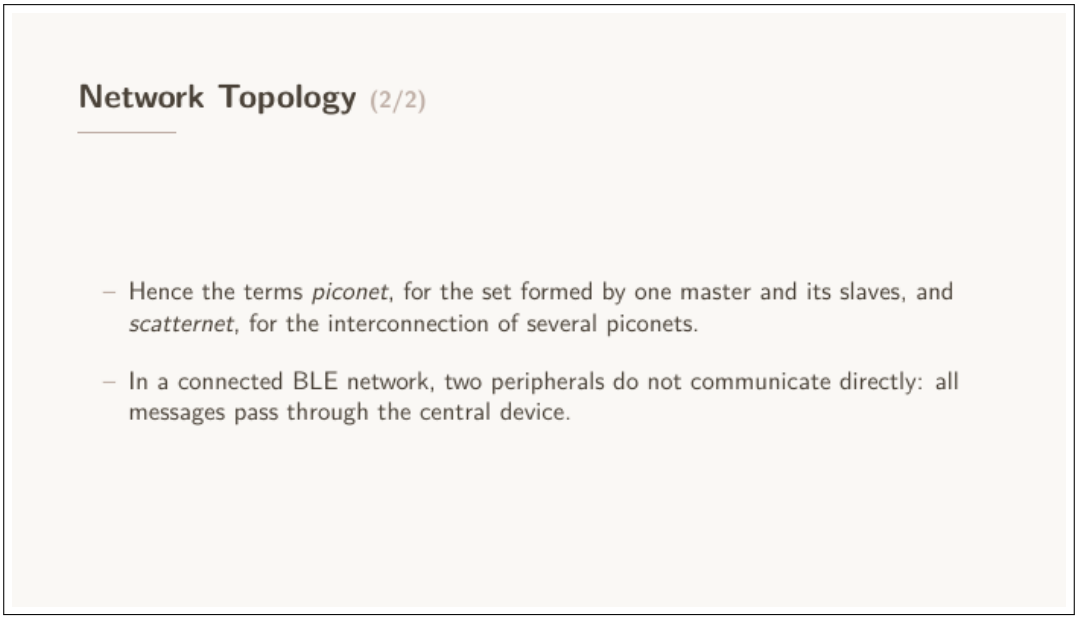
MTU, useful throughput and UUID. The default MTU is 23 bytes and leaves 20 useful bytes per read or notification. It can be negotiated up to 517 bytes. 16-bit UUIDs are abbreviations of the SIG base UUID, and multi-byte values follow little-endian. Check: what value is 2E 09 in little-endian?

Slide 42. Network Topology (1/2)



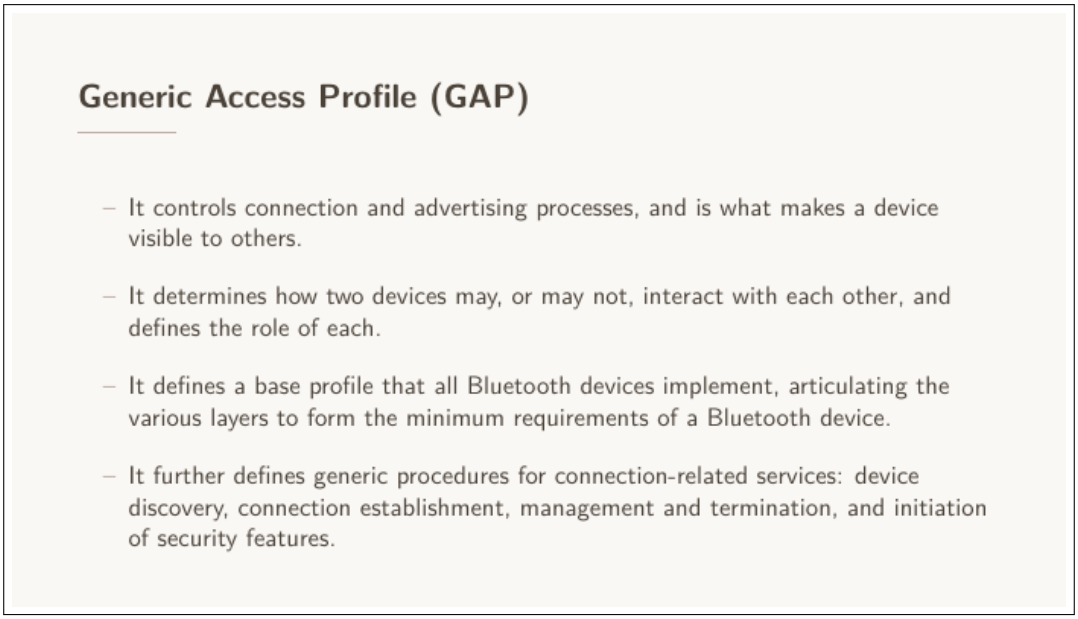
The star, in detail. The master manages the connection and may be connected to several slaves, each on a distinct physical channel. A slave only connects to one master at a time within the same connection. A device may, however, keep connections with more than one master, and play the roles of master and slave simultaneously.

Slide 43. Network Topology (2/2)



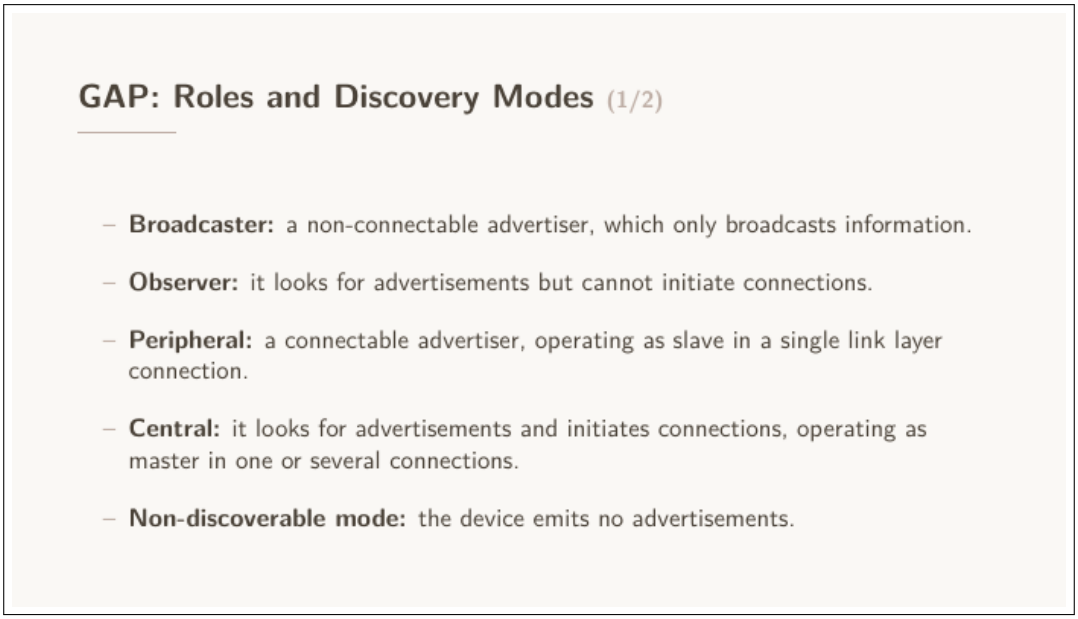
Piconet, scatternet and the limitation that matters. A piconet is the set of a master and its slaves. A scatternet is the interconnection of several piconets. The limitation to retain, already noted on slide 10, is that two peripherals do not communicate directly. Every message goes through the central.

Slide 44. Generic Access Profile (GAP)



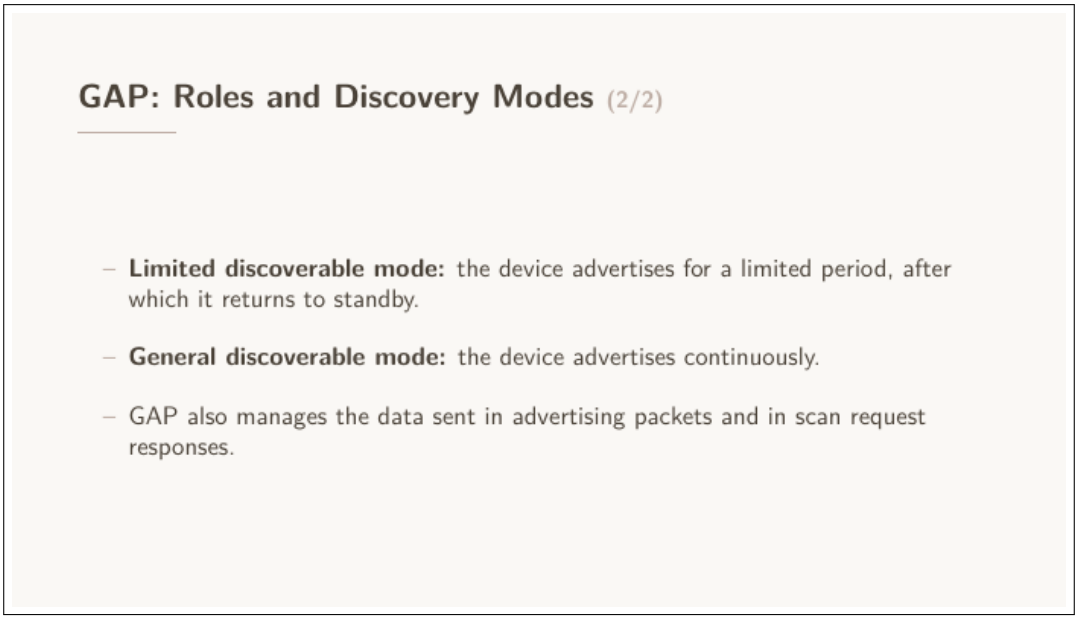
What GAP controls. GAP controls the connection and advertising processes, and is what makes a device visible. It determines how two devices can interact and defines the role of each. It also defines the generic procedures for discovery, establishment, management and termination of the connection, and the initiation of security features.

Slide 45. GAP: Roles and Discovery Modes (1/2)



The four GAP roles. Broadcaster advertises without accepting connections. Observer looks for advertisements without initiating any. Peripheral advertises and accepts a connection, operating as slave. Central looks for and initiates connections, operating as master. The first two correspond to connectionless communication; the last two, to connection-oriented communication.

Slide 46. GAP: Roles and Discovery Modes (2/2)



Discovery modes. Limited mode advertises for a defined period and then returns to rest. General mode advertises continuously. The choice between the two has privacy consequences, taken up in the activity band case study.

Slide 47. Security: Security Manager Protocol (1/2)



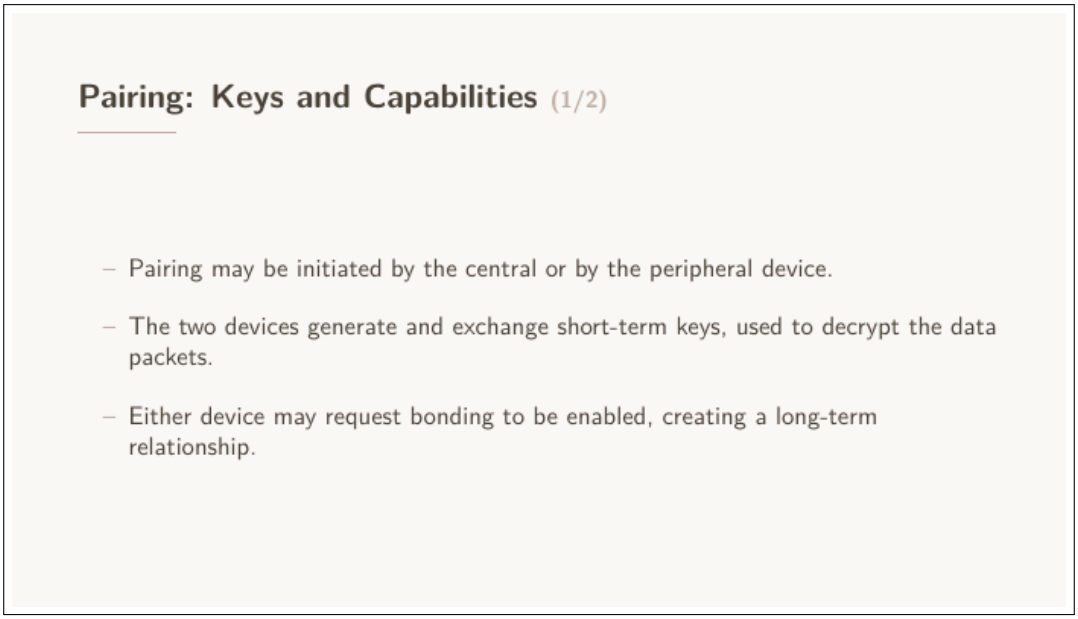
SMP and the three mechanisms. The Security Manager Protocol handles authentication and key management, using AES-128. It works with GAP through three distinct mechanisms, which should not be confused. Pairing establishes encryption between two devices once the connection is created.

Slide 48. Security: Security Manager Protocol (2/2)



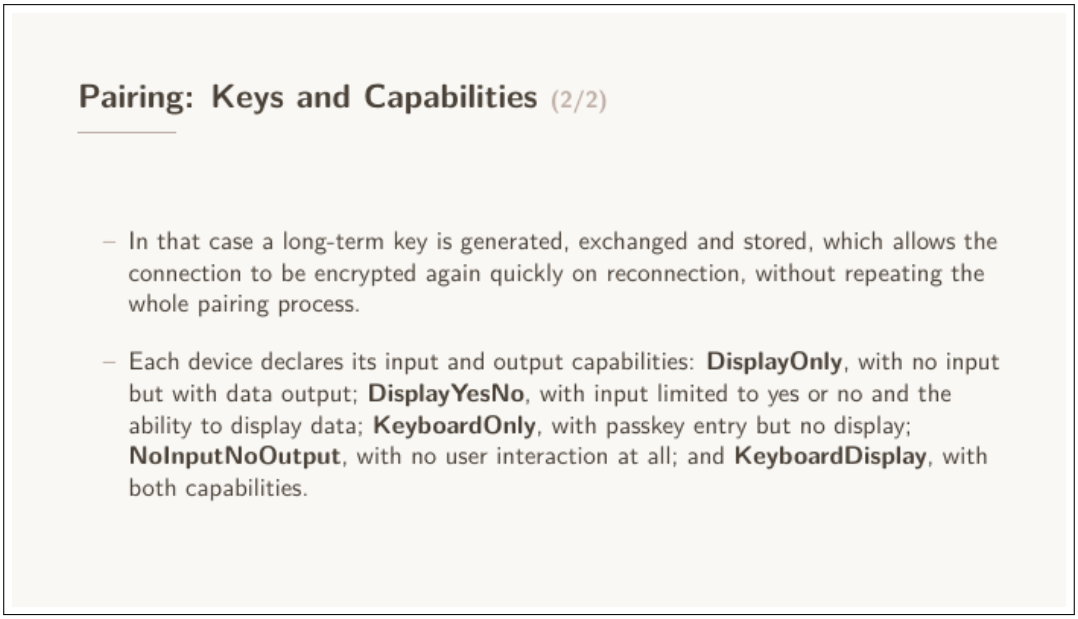
Authentication and bonding. Authentication verifies that the peer is trustworthy, protecting against man-in-the-middle attacks. Bonding establishes a long-term relationship, with storage of security and identity information. Encrypting is not the same as authenticating. This distinction is decisive on slide 44.

Slide 49. Pairing: Keys and Capabilities (1/2)



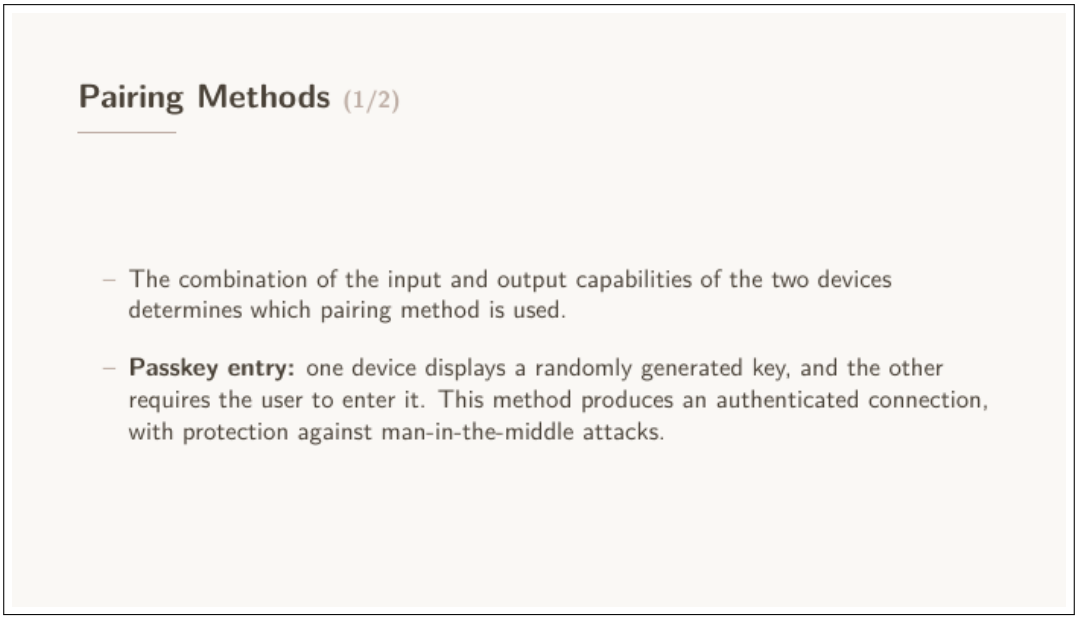
Short-term and long-term keys. In pairing, the devices generate and exchange short-term keys, used to decrypt the packets. Either party may ask for bonding to be activated.

Slide 50. Pairing: Keys and Capabilities (2/2)



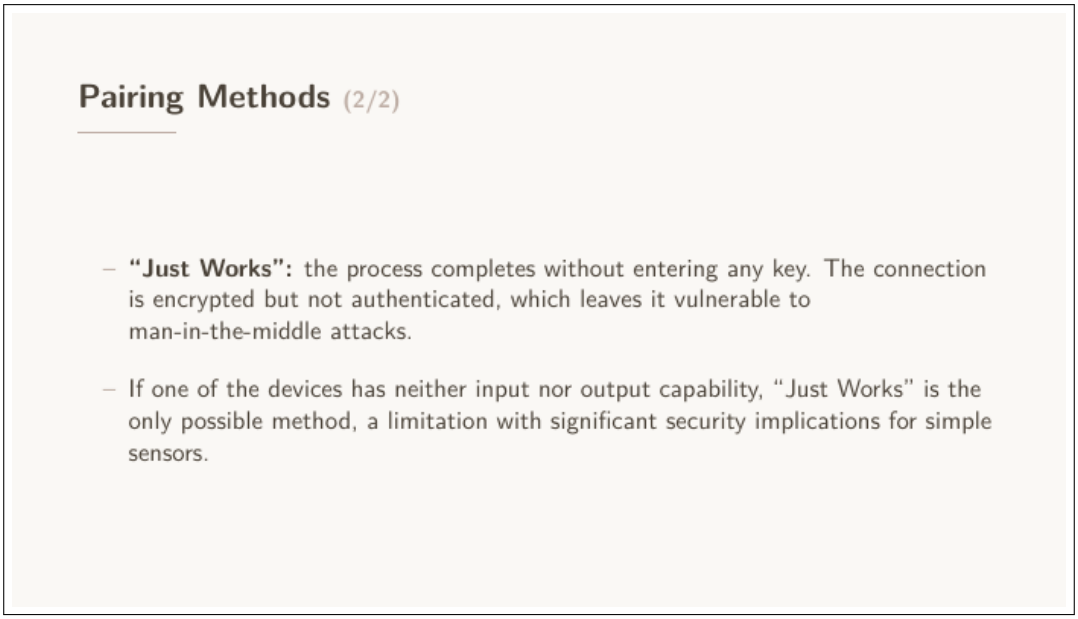
Input and output capabilities. Once bonding is activated, a long-term key is generated and stored, which speeds up reconnection without repeating the full pairing. The five capability categories, from DisplayOnly to KeyboardDisplay, should be retained, because they determine the pairing method available. NoInputNoOutput is the category into which most simple sensors fall.

Slide 51. Pairing Methods (1/2)



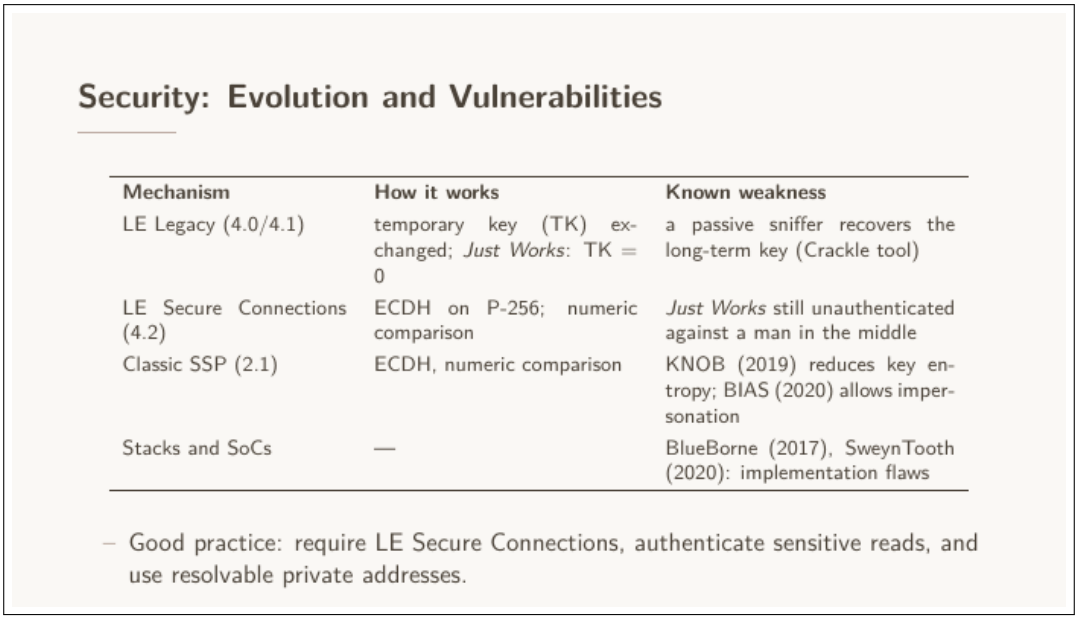
Passkey entry. The combination of the capabilities of the two devices determines the method used. In passkey entry, one device shows a random key and the other requires the user to enter it. This method produces an authenticated connection, with protection against man-in-the-middle attacks.

Slide 52. Pairing Methods (2/2)



The Just Works method and its consequence. In the Just Works method the process completes without a key being entered. The connection is encrypted but not authenticated, which leaves it vulnerable to man-in-the-middle attacks. The design consequence is serious: if one of the devices has neither input nor output, this is the only possible method. On a sensor with no display or keyboard, protection against intermediaries is not available, and that follows from the standard and not from poor implementation.

Slide 53. Security: Evolution and Vulnerabilities



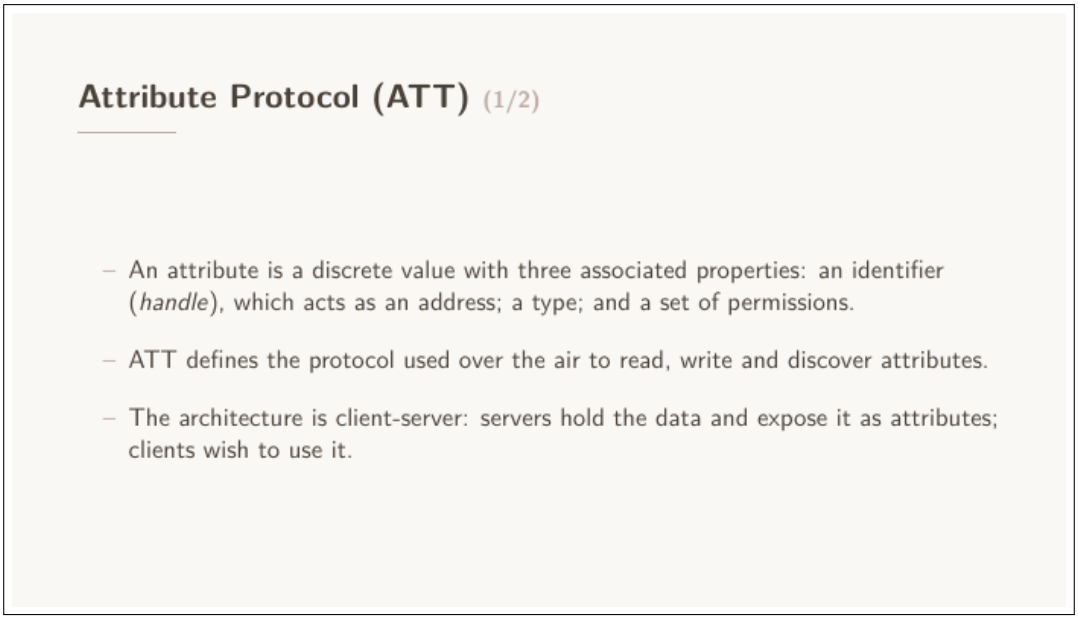
Security: Evolution and Vulnerabilities

Mechanism	How it works	Known weakness
LE Legacy (4.0/4.1)	temporary key (TK) exchanged; *Just Works*: TK = 0	a passive sniffer recovers the long-term key (Crackle tool)
LE Secure Connections (4.2)	ECDH on P-256; numeric comparison	*Just Works* still unauthenticated against a man in the middle
Classic SSP (2.1)	ECDH, numeric comparison	KNOB (2019) reduces key entropy; BIAS (2020) allows impersonation
Stacks and SoCs	—	BlueBorne (2017), SweynTooth (2020): implementation flaws

– Good practice: require LE Secure Connections, authenticate sensitive reads, and use resolvable private addresses.

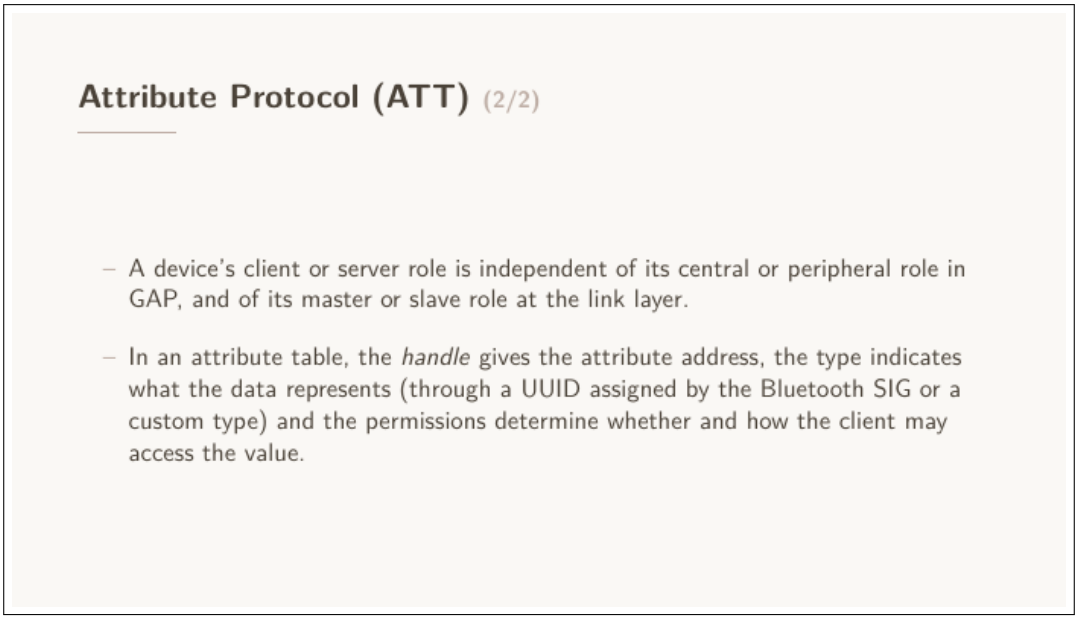
Security: evolution and vulnerabilities. The table compares the mechanisms and their known weaknesses. Legacy pairing with Just Works can be recovered by a passive sniffer. Secure Connections improves the cipher, but Just Works remains unauthenticated. There are also implementation flaws in the stacks. Good practices are to require Secure Connections, authenticate sensitive reads and use private addresses.

Slide 54. Attribute Protocol (ATT) (1/2)



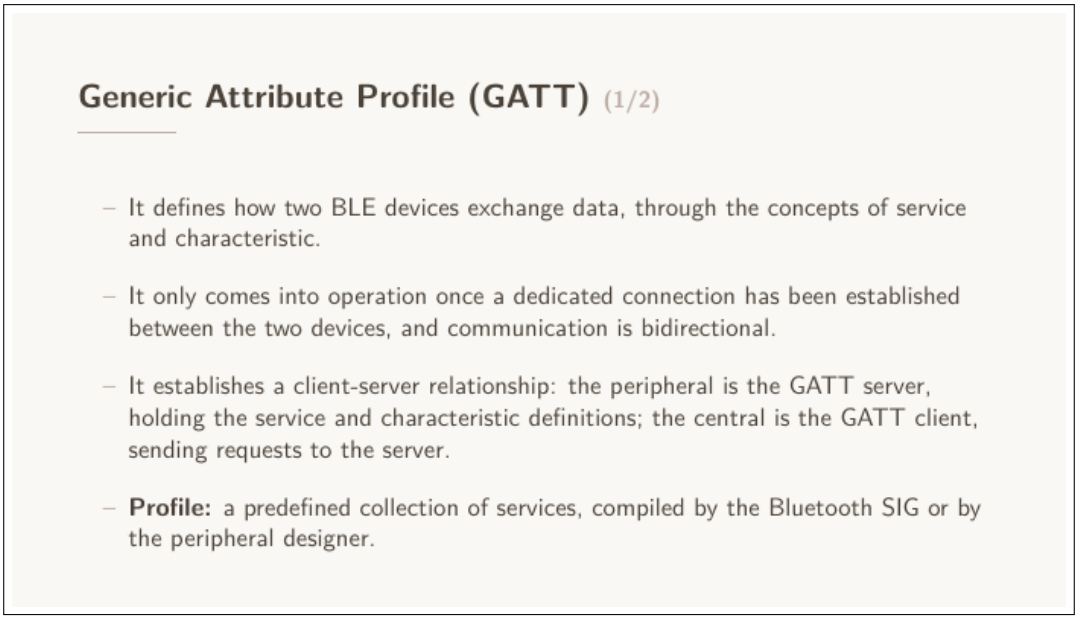
What an attribute is. An attribute is a discrete value with three properties: a handle, which works as an address, a type and a set of permissions. ATT defines the over-the-air protocol to read, write and discover attributes. The architecture is client-server: servers hold the data, clients want to use it.

Slide 55. Attribute Protocol (ATT) (2/2)



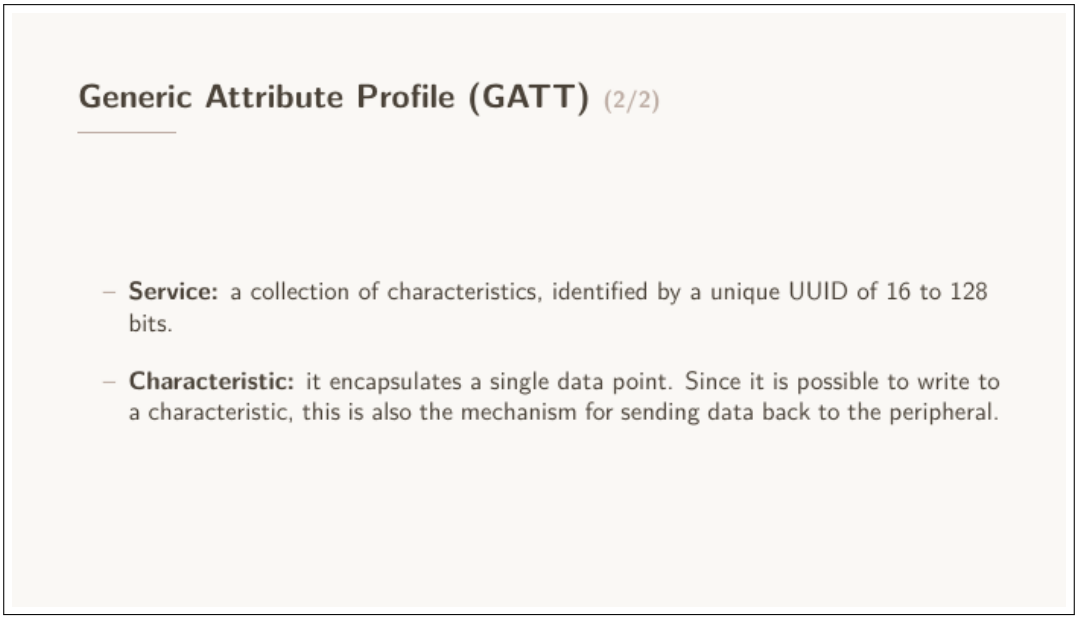
Independence of roles and reading the table. The client or server role is independent of the GAP role and of the role at the link layer. In an attribute table, the handle gives the address, the type indicates what the data represents through a UUID, and the permissions determine access. In the lab session this table is observed on a real device.

Slide 56. Generic Attribute Profile (GATT) (1/2)



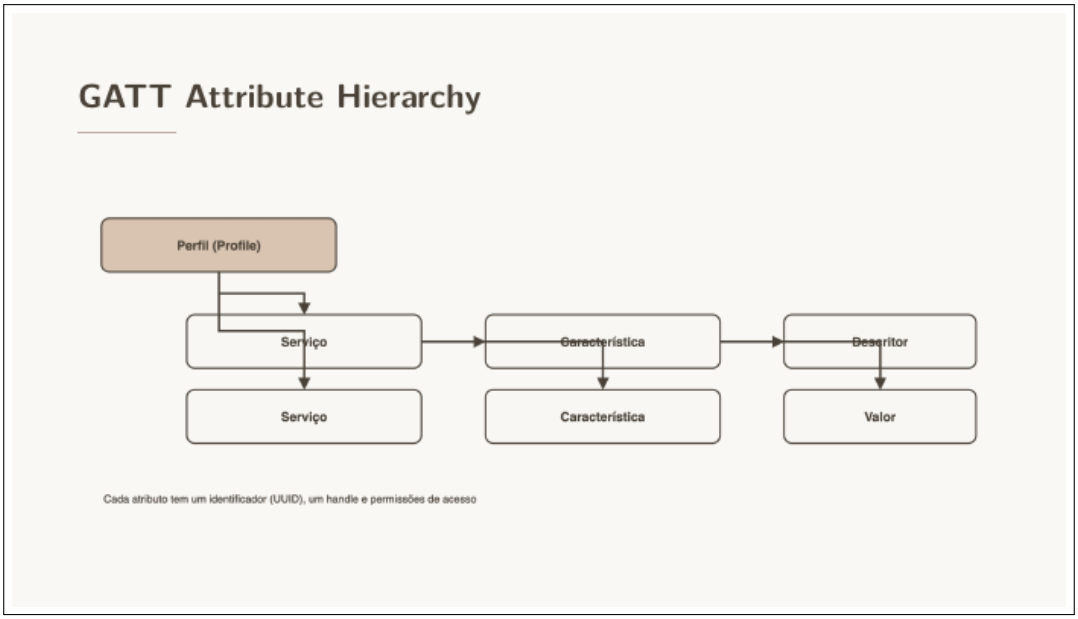
Services and characteristics. GATT defines how two devices exchange data, through the concepts of service and characteristic. It only comes into operation after a dedicated connection is established. The peripheral is the server, which holds the definitions; the central is the client, which sends requests.

Slide 57. Generic Attribute Profile (GATT) (2/2)



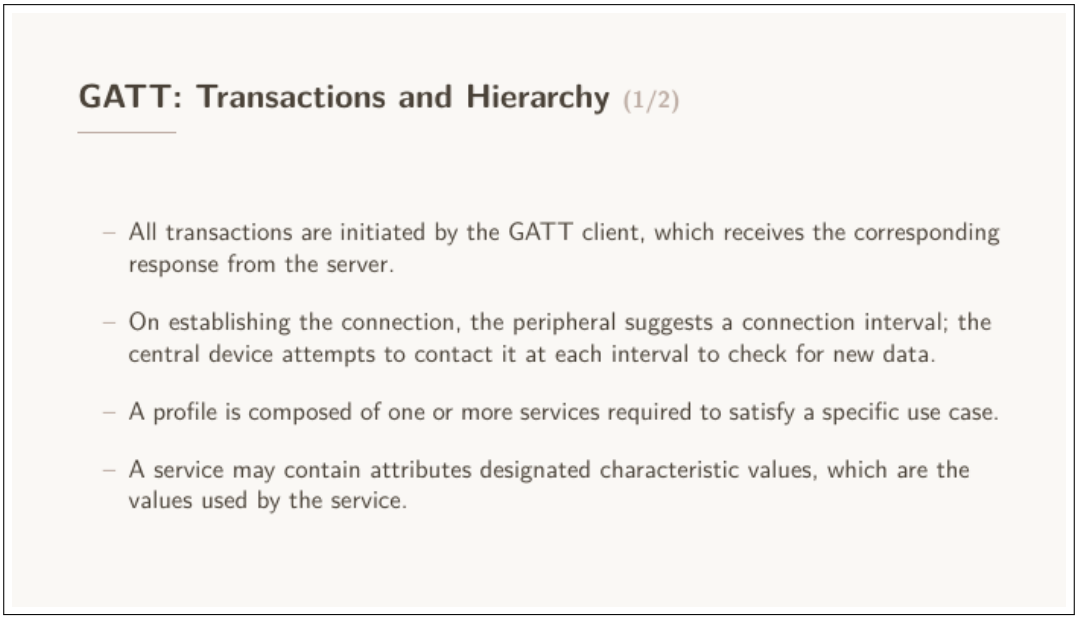
Profile, service and characteristic. A profile is a predefined collection of services. A service is a collection of characteristics, identified by a 16- to 128-bit UUID. A characteristic encapsulates a single data point. Note that it is possible to write to a characteristic: that is the mechanism for sending data to the peripheral.

Slide 58. GATT Attribute Hierarchy



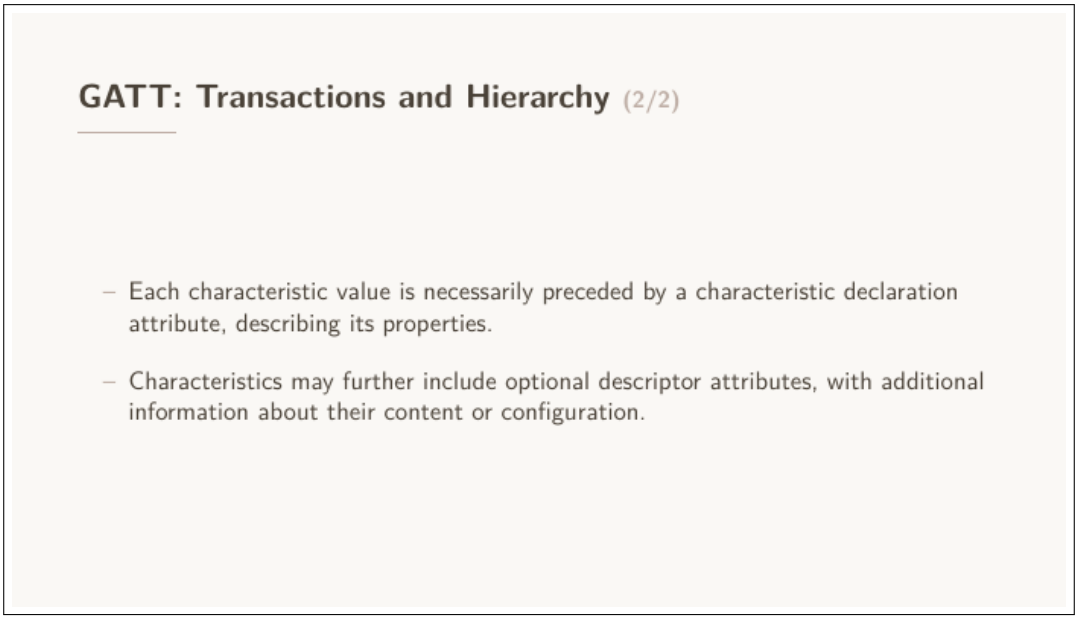
The complete hierarchy. The diagram shows the chain from profile to descriptor. Each attribute has a UUID, a handle and permissions. This hierarchy is the one walked in the lab session with the primary command, followed by characteristics.

Slide 59. GATT: Transactions and Hierarchy (1/2)



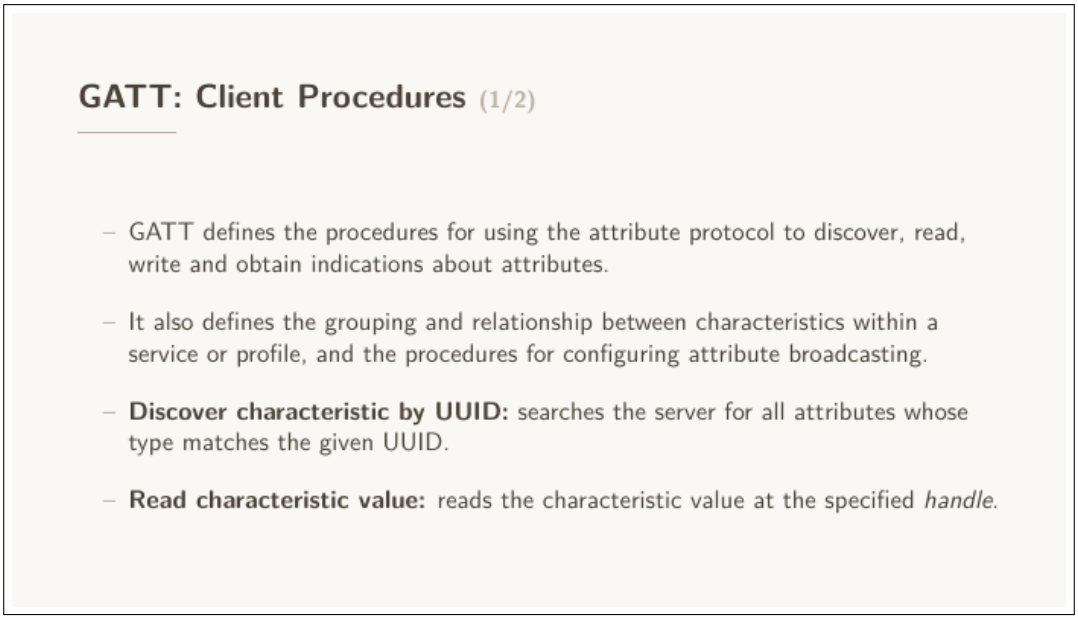
Who starts the transactions. All transactions are initiated by the client. The peripheral suggests a connection interval and the central contacts it at each interval to check whether there is new data. A profile is made up of the services needed to satisfy a concrete use case.

Slide 60. GATT: Transactions and Hierarchy (2/2)



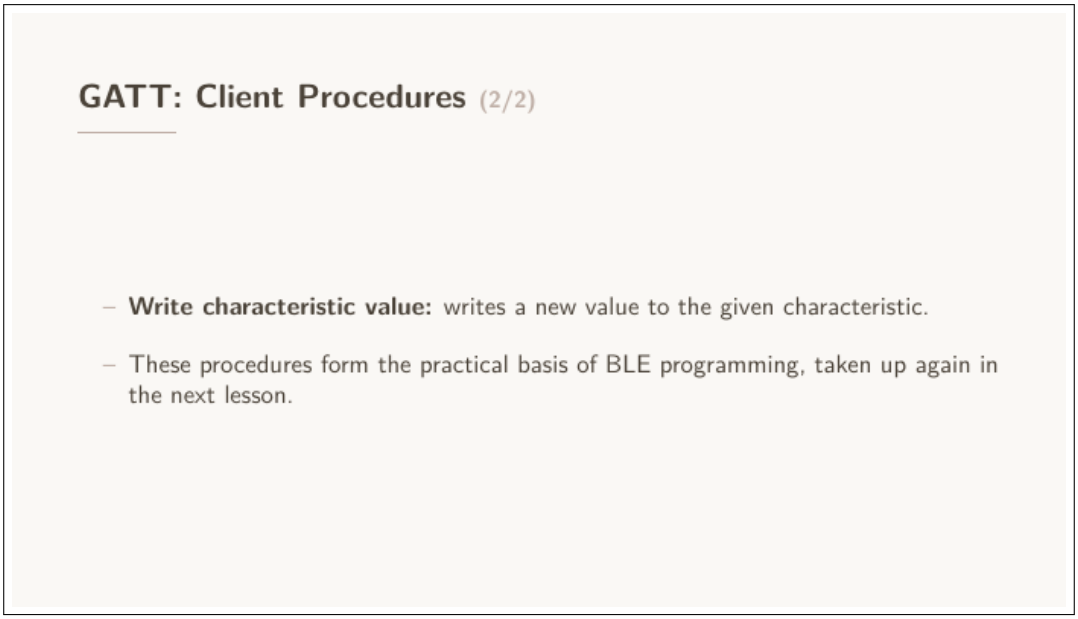
Declaration and descriptors. Each characteristic value is necessarily preceded by a declaration attribute, which describes its properties. Optional descriptors with additional information may exist. It is in a descriptor that notifications are enabled, the subject of the next lecture.

Slide 61. GATT: Client Procedures (1/2)



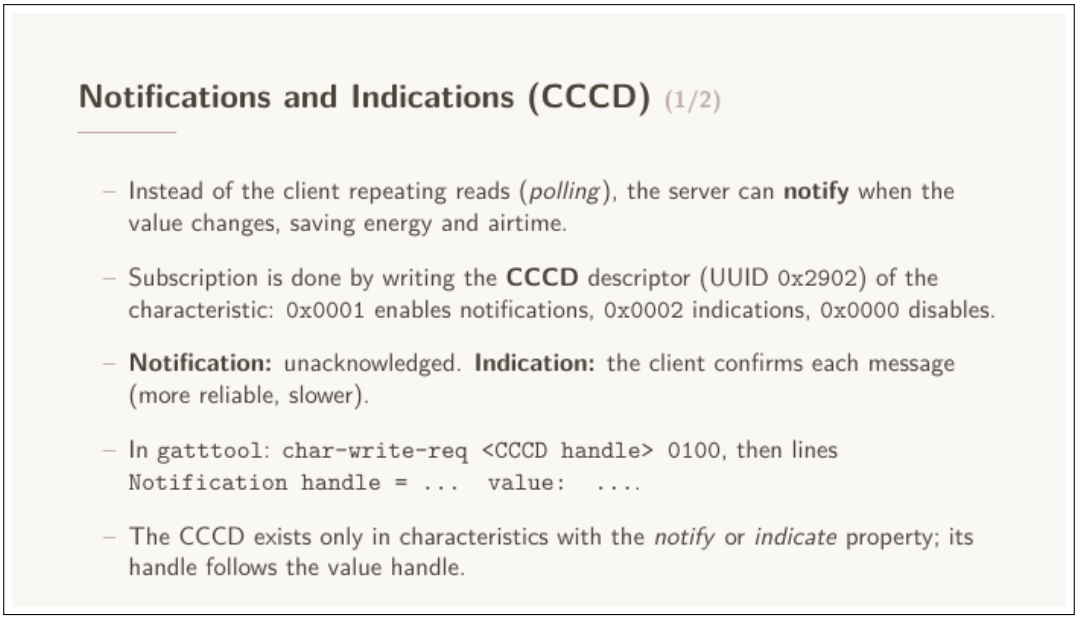
The procedures available. GATT defines the procedures to discover, read, write and obtain indications. Discover characteristic by UUID searches the server for all attributes of a given type. Read value obtains the content at the indicated handle.

Slide 62. GATT: Client Procedures (2/2)



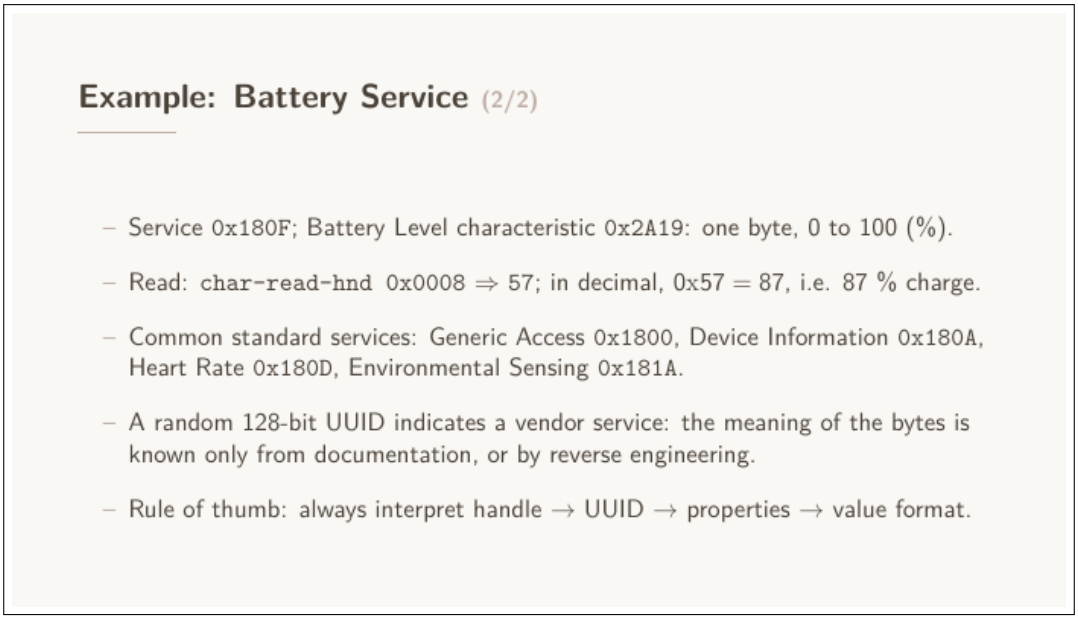
Writing and the link to the next lecture. Write value puts new content in the indicated characteristic. These procedures are the practical basis of BLE programming, resumed in Lecture 05.

Slide 63. Notifications and Indications (CCCD) (1/2)



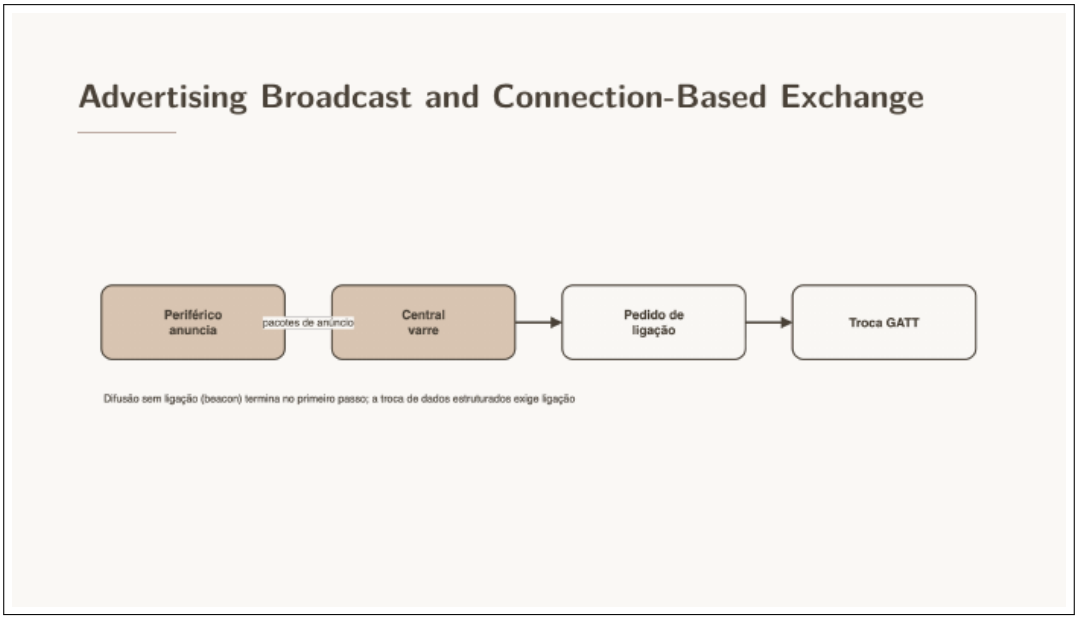
Notifications and indications. Instead of the client repeating reads, the server notifies when the value changes. Subscription is done by writing to the CCCD, with 0001 for notifications and 0002 for indications. An indication requires confirmation and is more reliable, but slower. Check: what value is written to enable notifications?

Slide 64. Example: Battery Service (2/2)



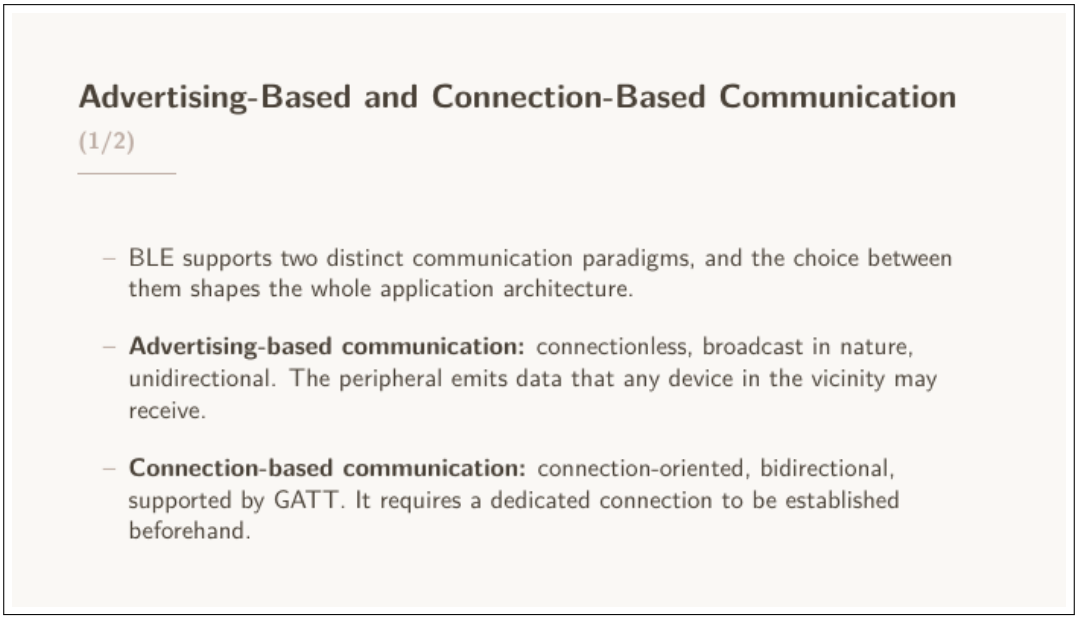
Example: the battery service. Service 0x180F and characteristic 0x2A19 return a byte between 0 and 100. A reading of 0x57 corresponds to 87 per cent. The practical rule is to always interpret the handle, the UUID, the properties and the value format, in that order.

Slide 65. Advertising Broadcast and Connection-Based Exchange



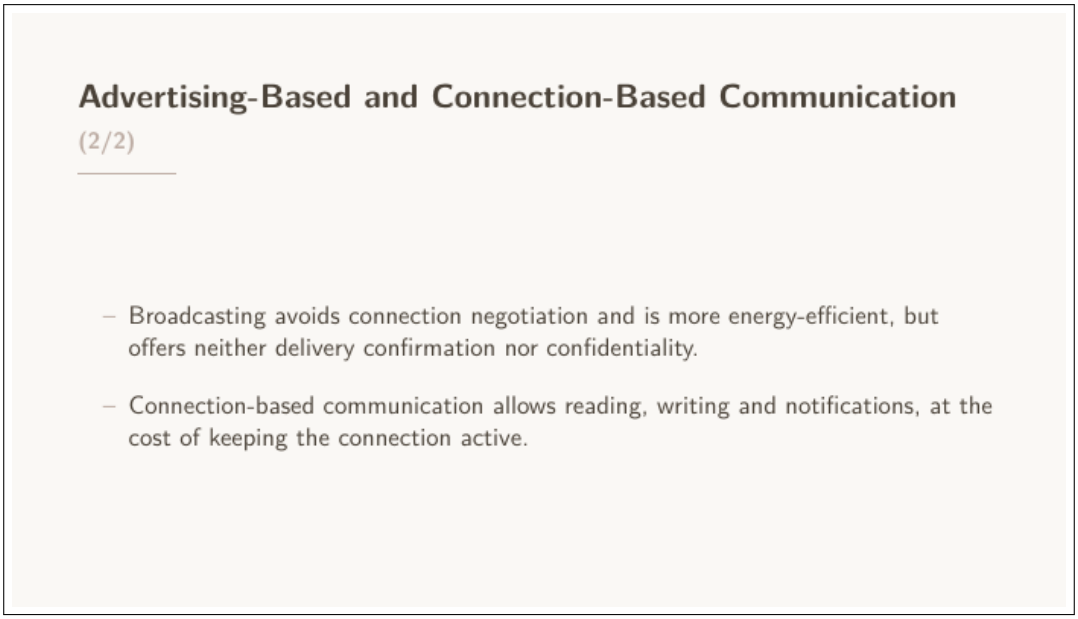
The two paradigms, side by side. The diagram shows the four steps of the communication. Broadcast advertising ends at the first step. The exchange of structured data requires going through all four.

Slide 66. Advertising-Based and Connection-Based Communication (1/2)



Broadcast and connection. Advertising-based communication is connectionless, unidirectional, of broadcast type. Connection-based communication is bidirectional and sustained by GATT. The choice between the two conditions the whole application architecture, and should be made early.

Slide 67. Advertising-Based and Connection-Based Communication (2/2)



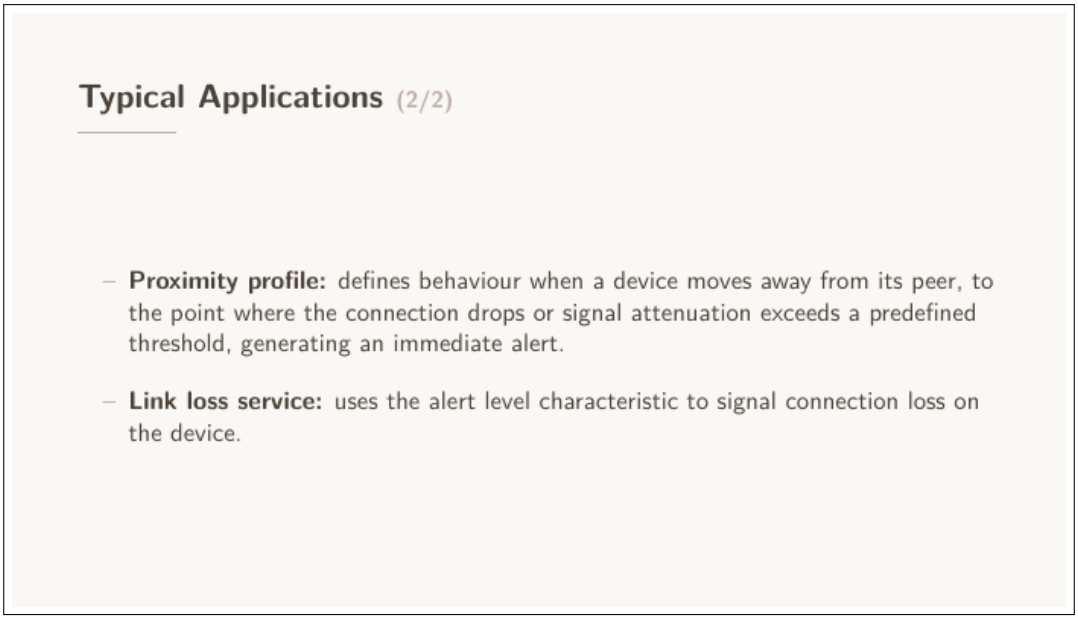
What is gained and what is lost. Broadcasting dispenses with negotiation and is cheaper, but offers neither delivery confirmation nor confidentiality. The connection allows reading, writing and notifications, at the cost of keeping it active.

Slide 68. Typical Applications (1/2)



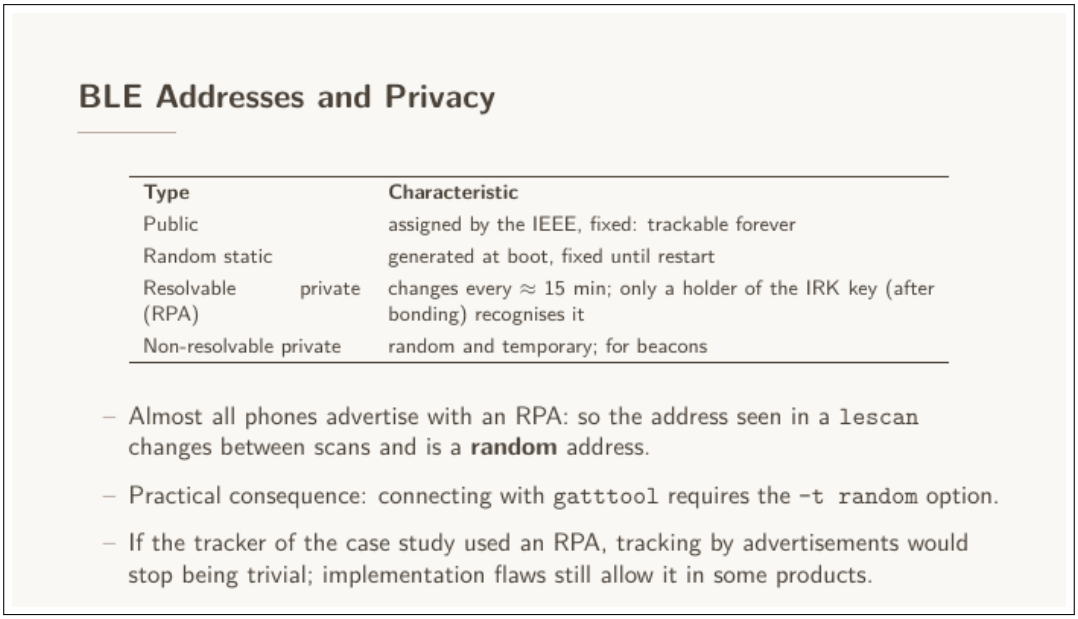
Where BLE applies. Low consumption and low duty cycles define the field of application. Objects we carry, objects around us, and communication inside systems.

Slide 69. Typical Applications (2/2)



Two standardised profiles. The proximity profile defines the behaviour when a device moves away from its peer to the point where the connection drops or the attenuation exceeds a threshold. The link loss service signals that loss through the alert level characteristic.

Slide 70. BLE Addresses and Privacy



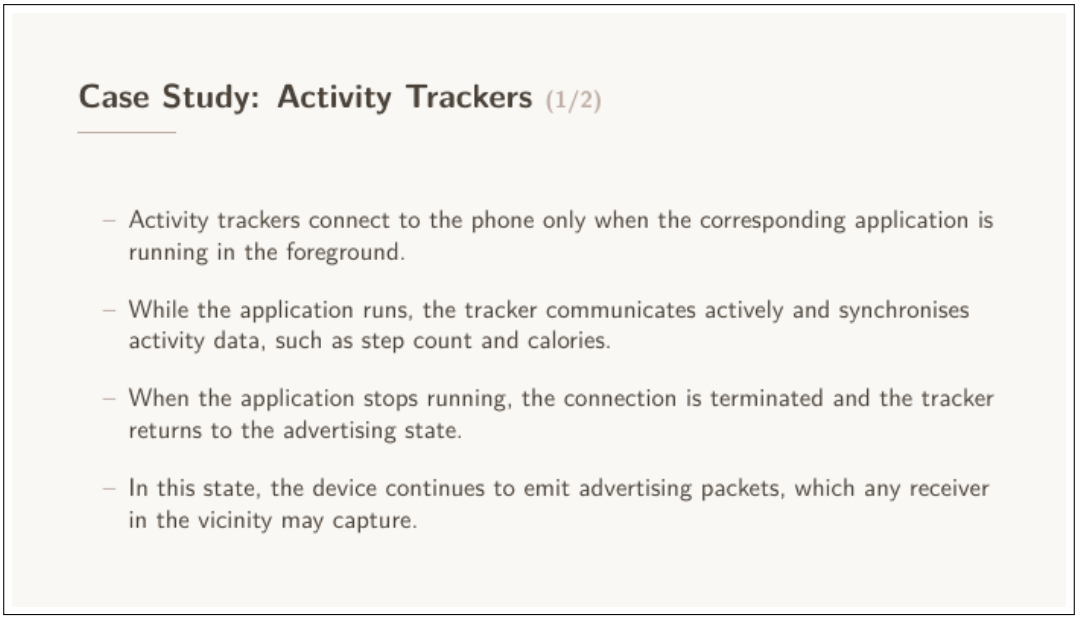
BLE Addresses and Privacy

Type	Characteristic
Public	assigned by the IEEE, fixed: trackable forever
Random static	generated at boot, fixed until restart
Resolvable private (RPA)	changes every ≈ 15 min; only a holder of the IRK key (after bonding) recognises it
Non-resolvable private	random and temporary; for beacons

- Almost all phones advertise with an RPA: so the address seen in a `lescan` changes between scans and is a **random** address.
- Practical consequence: connecting with `gatttool` requires the `-t random` option.
- If the tracker of the case study used an RPA, tracking by advertisements would stop being trivial; implementation flaws still allow it in some products.

BLE addresses and privacy. Most phones use resolvable private addresses, which change, and so connecting with gatttool requires the random address option. The same mechanism makes tracking by advertisements harder, and takes up the activity band case study again: with private addresses, the leak would be far less trivial.

Slide 71. Case Study: Activity Trackers (1/2)



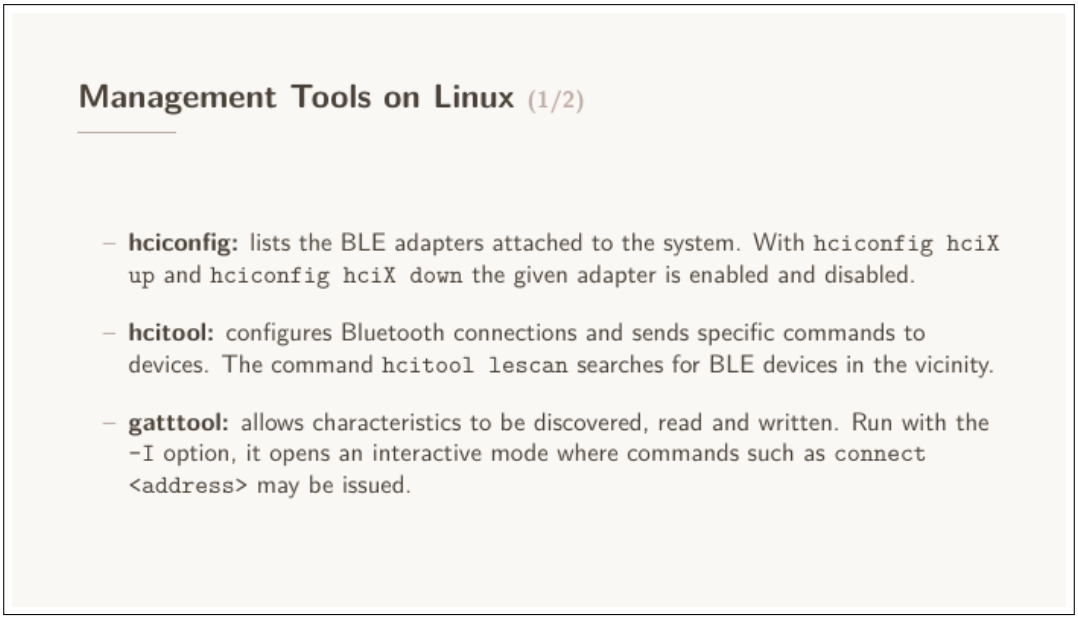
How activity bands work. The band connects to the phone only when the app runs in the foreground. While that happens, it synchronises the activity data. When the app stops, the connection ends and the band returns to the advertising state. In that state it keeps emitting packets that any receiver can pick up.

Slide 72. Case Study: Activity Trackers (2/2)



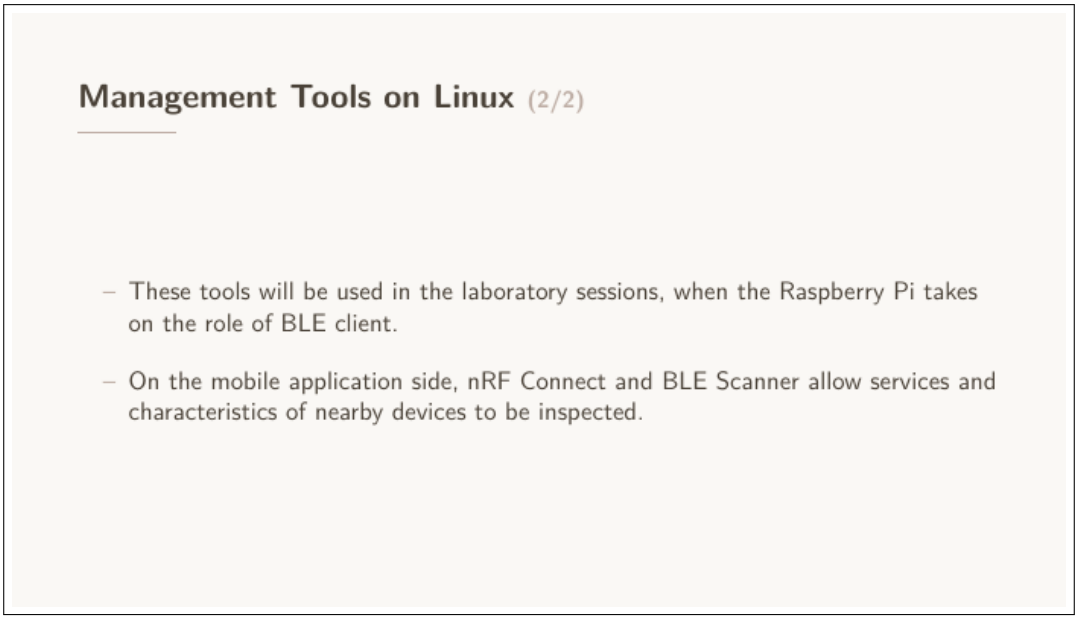
The privacy leak. From the continuous emission of advertisements comes the possibility of tracking the user's presence and movement, without any connection being established and without their knowledge. The example is important because it shows that decisions taken at protocol level, for legitimate operational reasons, have direct privacy consequences.

Slide 73. Management Tools on Linux (1/2)



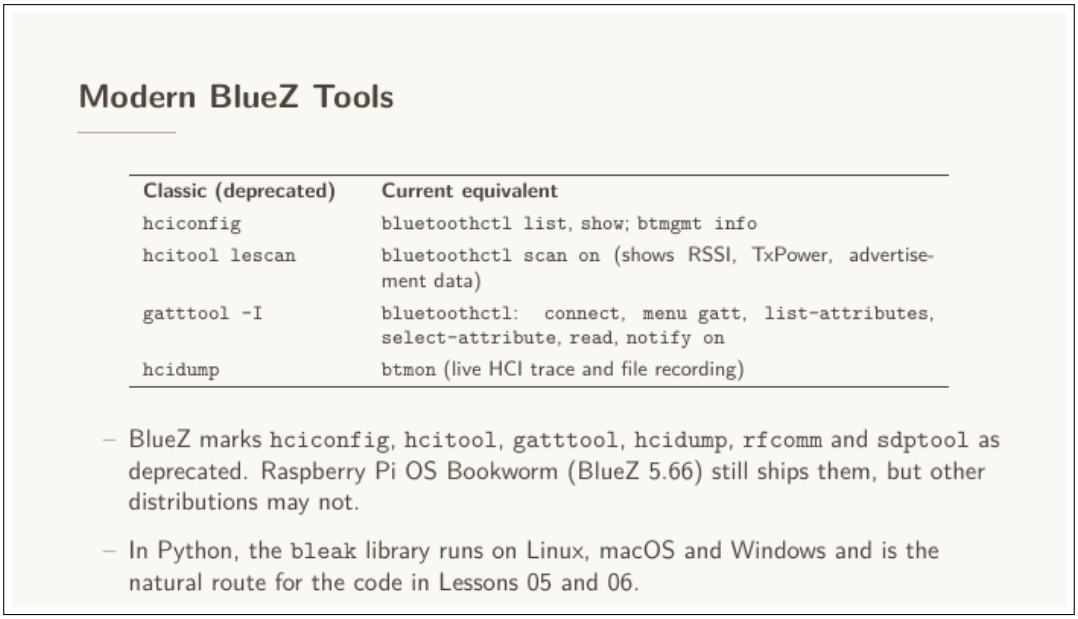
The three tools. hciconfig lists and enables adapters. hcitool configures connections and looks for devices with the lescan subcommand. gatttool discovers, reads and writes characteristics, and with the -I option opens an interactive mode. These commands are used directly in the lab session.

Slide 74. Management Tools on Linux (2/2)



On the gateway and phone side. The tools will be used when the Raspberry Pi takes the role of BLE client, in the following lectures. On the mobile side, nRF Connect and BLE Scanner let services and characteristics be inspected.

Slide 75. Modern BlueZ Tools



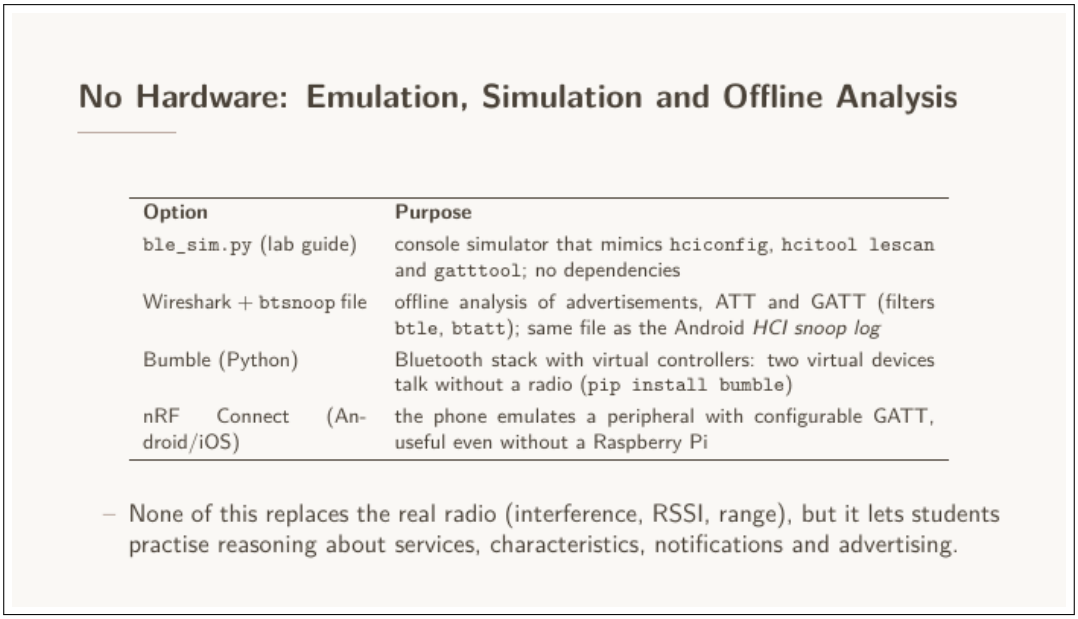
Modern BlueZ Tools

Classic (deprecated)	Current equivalent
`hciconfig`	`bluetoothctl list`, `show`; `btmgmt info`
`hcitool lescan`	`bluetoothctl scan on` (shows RSSI, TxPower, advertisement data)
`gatttool -I`	`bluetoothctl`: `connect`, `menu gatt`, `list-attributes`, `select-attribute`, `read`, `notify on`
`hcidump`	`btmon` (live HCI trace and file recording)

- BlueZ marks `hciconfig`, `hcitool`, `gatttool`, `hcidump`, `rfcomm` and `sdptool` as deprecated. Raspberry Pi OS Bookworm (BlueZ 5.66) still ships them, but other distributions may not.
- In Python, the `bleak` library runs on Linux, macOS and Windows and is the natural route for the code in Lessons 05 and 06.

Modern BlueZ tools. The classic BlueZ tools are marked as deprecated. Raspberry Pi OS Bookworm still includes them, but other distributions may lack them. The Python alternative is the bleak library, which works on Linux, macOS and Windows and is the basis of the code in Lectures 05 and 06.

Slide 76. No Hardware: Emulation, Simulation and Offline Analysis



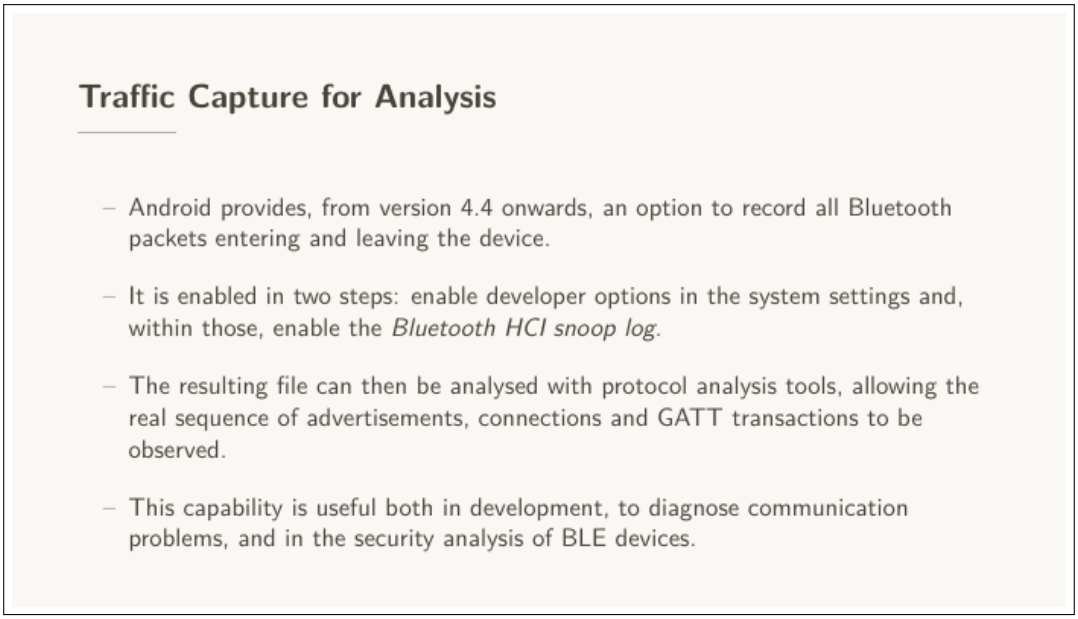
No Hardware: Emulation, Simulation and Offline Analysis

Option	Purpose
`ble_sim.py` (lab guide)	console simulator that mimics `hciconfig`, `hcitool lescan` and `gatttool`; no dependencies
Wireshark + `btsnoop` file	offline analysis of advertisements, ATT and GATT (filters `btle`, `btatt`); same file as the Android *HCI snoop log*
Bumble (Python)	Bluetooth stack with virtual controllers: two virtual devices talk without a radio (`pip install bumble`)
nRF Connect (Android/iOS)	the phone emulates a peripheral with configurable GATT, useful even without a Raspberry Pi

– None of this replaces the real radio (interference, RSSI, range), but it lets students practise reasoning about services, characteristics, notifications and advertising.

Without hardware. The table lists ways to exercise the reasoning without a radio: a console simulator, offline analysis in Wireshark, the Bumble virtual stack and peripheral emulation on the phone. None replaces the real radio, but they let you practise services, characteristics, notifications and advertisements.

Slide 77. Traffic Capture for Analysis



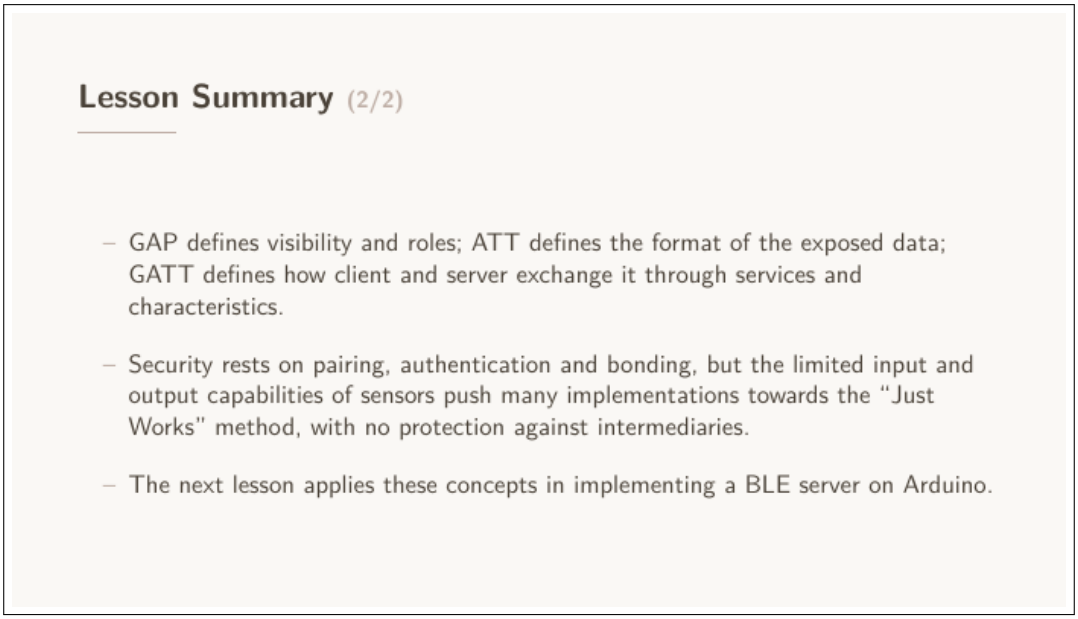
Observing the protocol at work. Android allows, since version 4.4, all Bluetooth packets entering and leaving the device to be logged. Activation is done in two steps, through the developer options. The generated file can be analysed with protocol analysis tools, which lets the real sequence of advertisements, connections and GATT transactions be observed. It is useful in development and in security analysis.

Slide 78. Lesson Summary (1/2)



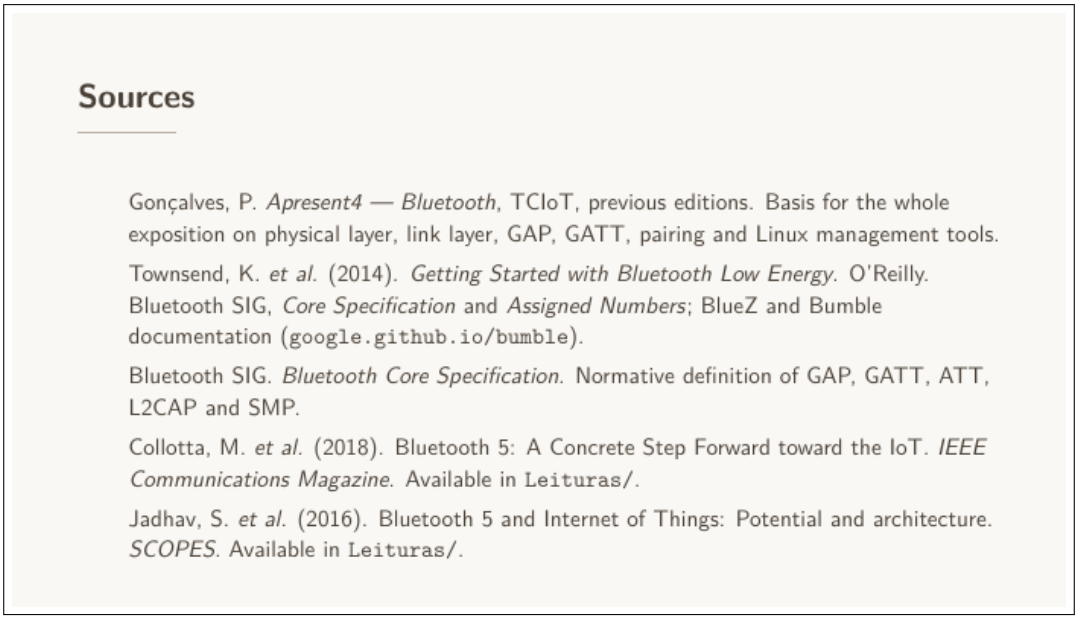
Summary, first part. Classic Bluetooth and BLE share a brand and a band, but are distinct, non-interoperable technologies. The link layer organises the whole communication around six states and two modes. The connection parameters are the main instrument for balancing autonomy and responsiveness.

Slide 79. Lesson Summary (2/2)



Summary, second part. GAP defines visibility and roles, ATT defines the format of the data exposed, GATT defines how client and server exchange them. The conclusion on security deserves a note: the limited capabilities of sensors push many implementations to the Just Works method, with no protection against intermediaries. It is not an implementation defect, it is a consequence of the standard.

Slide 80. Sources



Origin of the material. The basis is the course material from previous editions, complemented by the Bluetooth SIG specification. The two papers on Bluetooth 5, available in the readings folder, cover the extensions of range and broadcast capacity after version 4.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Distinguish the three pairs of roles (master and slave, central and peripheral, client and server) and state the layer of each.
2. Explain why BLE reaches a greater distance than classic Bluetooth, despite its lower consumption.
3. Describe the path of a device through the six link-layer states, from discovery to termination.
4. State the effect of increasing the connection interval and of increasing slave latency.
5. Explain why a sensor with no display or keyboard cannot establish an authenticated connection.
6. Walk the GATT hierarchy, from profile to descriptor, stating what each level identifies.
7. Justify the choice between advertising-based and connection-based communication for a given use case.
8. Explain the privacy leak of activity bands and state in which protocol state it occurs.

References

- Gonçalves, P. *Apresent4 — Bluetooth*, TCIoT, previous editions.
- Bluetooth SIG. *Bluetooth Core Specification*.
- Collotta, M. *et al.* (2018). Bluetooth 5: A Concrete Step Forward toward the IoT. *IEEE Communications Magazine*.
- Jadhav, S. *et al.* (2016). Bluetooth 5 and Internet of Things: Potential and architecture. *SCOPES*.